Two-Timescale Secure Transmission for Movable Antenna-Enabled Multiuser Systems With Statistical Eavesdropper CSI

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***Abstract*—This paper investigates two-timescale secure transmission for a movable antenna (MA)-enabled multiuser downlink with a passive eavesdropper (Eve), where the antenna positions are optimized based on statistical channel state information (CSI), and zero-forcing precoding with null-space artificial noise (AN) adapts to the instantaneous CSI of the legitimate users (LUs). Specifically, the antenna positions and the long-term power allocation of the confidential signals and AN are jointly optimized to maximize the ergodic secrecy sum rate (ESSR), subject to the transmit power, movable-region, and minimum-distance constraints. To solve this nonconvex stochastic problem, we first transform it into a deterministic problem by approximating the ergodic wiretap rate in closed form. The resulting approximation reveals that the MAs can reduce the information leakage and increase the AN received by Eve, even for a single LU. Then, an alternating optimization algorithm is developed for the deterministic problem, where the power allocation among the LUs follows a secrecy water-filling structure, and each antenna position is selected by a grid search and refined by majorization–minimization. Simulation results show that the proposed design outperforms the benchmark schemes while allocating less power to the AN, and requires fewer antennas to achieve a given ESSR.**

***Index Terms*—Artificial noise, movable antenna, physical layer security, statistical CSI, two-timescale design.**

I. Introduction

PHYSICAL layer security (PLS) exploits the randomness and spatial characteristics of wireless channels to protect confidential messages against eavesdropping [1]–[3]. In multi-antenna systems, PLS is commonly achieved by secure precoding and artificial noise (AN) [4], [5], which steer the confidential signals toward the legitimate users (LUs) and degrade the channel of the eavesdropper (Eve), respectively. Both techniques, however, rely on the spatial separability between the LUs and Eve [2]. This separability is limited for a conventional fixed-position antenna (FPA) array if Eve is located in a direction close to that of an LU and the line-of-sight (LoS) path is dominant. Recently, movable antennas (MAs) [6], [7] have emerged as a means to enhance this separability, where adjusting the antenna positions within a given region reshapes the channel correlation among different receivers without additional radio-frequency (RF) chains [8]–[10].

For MA-enabled PLS, the transmit beamforming and the antenna positions have been jointly optimized to maximize the secrecy rate with perfect channel state information (CSI) [11] and without the instantaneous CSI of the eavesdroppers [12]. A deterministic equivalent of the ergodic secrecy rate has been derived for a multiple-input multiple-output MA system with the statistical CSI of a multi-antenna Eve [13]. In [14], a sensing-assisted secure design has been developed, where the antenna positions improve both the estimation of the angles of departure (AoDs) of Eve and the secrecy rate. However, the antenna positions in [11]–[13] are optimized based on the instantaneous CSI of the legitimate receivers. Since this CSI varies from one channel coherence interval (CI) to another, the antennas have to be repositioned in each CI, which incurs a high operational complexity [15]. These designs also require the CSI over the entire movable region, whose acquisition entails channel measurements at many antenna positions [15], [16]. The design in [14], on the other hand, assumes LoS channels and thus ignores the small-scale fading.

To reduce the frequency of antenna movement and the CSI acquisition overhead, two-timescale designs have been developed for MA systems [15], [16], following those for intelligent reflecting surfaces [17]. In these designs, the antenna positions are optimized based on the statistical CSI, whereas the beamforming is designed based on the instantaneous CSI at the optimized positions. Specifically, the antenna positions have been optimized for a multiuser downlink with maximum-ratio transmission and zero-forcing (ZF) precoding in [15] and for a multiuser uplink with linear and successive interference cancellation receivers in [16]. However, these works aim at maximizing the ergodic sum rate without considering eavesdropping. In particular, the ergodic-rate approximations for ZF precoding adopted in [15], [16], [18] characterize only the signal power received by the intended users. These approximations therefore depend on the antenna positions only through the channel correlations among the users, so that the MAs bring no rate gain for a single user [15, Remark 2]. For secure transmission, however, the ergodic rate of Eve further depends on the correlations between the channels of Eve and the LUs, which cannot be captured by these approximations. To the best of our knowledge, the two-timescale MA design for secure transmission remains unexplored.

In this paper, we consider an MA-enabled multiuser downlink with a passive Eve, where the base station (BS) adopts ZF precoding and null-space AN. The antenna positions and the long-term power allocation are jointly optimized based on the statistical CSI to maximize the ergodic secrecy sum rate (ESSR), which requires neither the instantaneous CSI of Eve nor antenna movement in each CI. Since the ZF directions and

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the AN subspace vary with the random channels of the LUs, we first prove that the information leakage and the AN power received by Eve are conditionally independent given these channels. Based on this property, we derive a closed-form approximation of the ergodic wiretap rate, which captures the LoS correlation between Eve and the LUs. This approximation reveals that the MAs can reduce the information leakage and increase the AN received by Eve, even for a single LU.

Then, we develop an alternating optimization (AO) algorithm for the resulting deterministic problem. For given antenna positions and AN power, the optimal power allocation among the LUs is derived in a secrecy water-filling form, where an LU is allocated nonzero power only if its effective channel gain exceeds that of Eve for its signal. For a given power allocation, each antenna position is first selected by a grid search and then refined by majorization–minimization (MM), where a rank-one structure of the objective function reduces the cost of evaluating each grid point. Simulation results show that the proposed design outperforms the benchmark schemes, including the rate-oriented MA design and antenna selection (AS), while allocating less power to the AN, and that its gain persists for a single LU.

Notations: Scalars, vectors, and matrices are denoted by italic, bold lower-case, and bold upper-case letters, respectively. $\mathbb{C}^{a\times b}$ and $\mathbb{R}^{a\times b}$ denote the sets of $a\times b$ complex and real matrices, respectively. $(\cdot)^{\mathrm{T}}$, $(\cdot)^*$, and $(\cdot)^{\mathrm{H}}$ denote the transpose, conjugate, and conjugate transpose, and $|\cdot|$, $\|\cdot\|$, and $\mathrm{tr}(\cdot)$ the modulus (or the cardinality of a set), Euclidean norm, and trace, respectively. $[\mathbf{x}]_i$ and $[\mathbf{X}]_{i,j}$ denote the i-th entry of $\mathbf{x}$ and the (i,j)-th entry of $\mathbf{X}$, $\mathrm{diag}(x_1,\ldots,x_M)$ the diagonal matrix with diagonal entries $x_1,\ldots,x_M$, and $\mathbf{X}^{1/2}$ the entrywise square root of a nonnegative diagonal matrix $\mathbf{X}$. $\mathbf{I}_N$ denotes the $N\times N$ identity matrix, $\mathbf{e}_m$ the m-th column of an identity matrix, and $\mathcal{CN}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ the proper complex Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\mathbf{X}\succeq\mathbf{0}$ ($\mathbf{X}\succ\mathbf{0}$) means that $\mathbf{X}$ is positive semidefinite (definite), and $\mathbf{X}_1\succeq\mathbf{X}_2$ means $\mathbf{X}_1-\mathbf{X}_2\succeq\mathbf{0}$. $j\triangleq\sqrt{-1}$, $\mathbb{E}\{\cdot\}$ and $\mathrm{Var}\{\cdot\}$ denote the expectation and variance, $\mathrm{Re}\{\cdot\}$ the real part, $\nabla_{\mathbf{x}}f$ the gradient of f with respect to $\mathbf{x}$, $[x]^+\triangleq\max\{x,0\}$, $\emptyset$ the empty set, $\setminus$ the set difference, and $\mathcal{O}(\cdot)$ and $o(\cdot)$ the big-O and little-o notations, respectively.

II. System Model and Problem Formulation

A. MA-Enabled Secure Multiuser Downlink System

As illustrated in Fig. 1, we consider the downlink of a multiuser system, where a BS equipped with N MAs serves M single-antenna LUs in the presence of a single-antenna passive Eve. Let $\mathcal{M}\triangleq\{1,\ldots,M\}$ and $\mathcal{N}\triangleq\{1,\ldots,N\}$ denote the index sets of the LUs and the MAs, respectively. To transmit AN in the null space of the channels of the LUs, we assume $N\geq M+1$. The MAs are connected to the RF chains via flexible cables and can move within a two-dimensional square region $\mathcal{C}\triangleq[-A/2,A/2]\times[-A/2,A/2]$ of side length A [6], [7]. In the local Cartesian coordinate system of $\mathcal{C}$, the position of the n-th MA is denoted by $\mathbf{t}_n\triangleq[x_n,y_n]^{\mathrm{T}}\in\mathcal{C}$, and the positions of all the MAs are collected in $\mathbf{t}\triangleq[\mathbf{t}_1^{\mathrm{T}},\ldots,\mathbf{t}_N^{\mathrm{T}}]^{\mathrm{T}}\in\mathbb{R}^{2N\times1}$. To mitigate

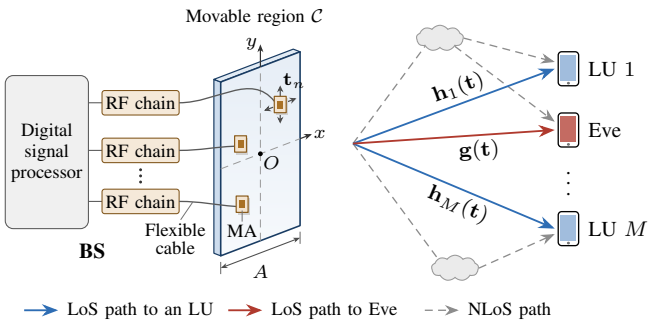


Fig. 1. An MA-enabled multiuser downlink system with a passive Eve.

mutual coupling, any two MAs must be separated by at least a minimum distance $D_{\min}$, and the residual coupling is neglected.

Since the size of $\mathcal{C}$ is much smaller than the propagation distance, moving the MAs changes only the phases of the LoS paths, whereas their AoDs and amplitudes remain unchanged [7]. Let $\theta_m\in[-\pi/2,\pi/2]$ and $\phi_m\in[-\pi/2,\pi/2]$ denote the elevation and azimuth AoDs of the LoS path from the BS to the m-th LU, respectively, and define $\mathbf{a}_m\triangleq[\cos\theta_m\sin\phi_m,\ \sin\theta_m]^{\mathrm{T}}$. Then, the propagation distance of the LoS path from the n-th MA differs from that from the origin of $\mathcal{C}$ by $\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_m$ [7], which results in a phase difference of $2\pi\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_m/\lambda$, where λ is the carrier wavelength. Hence, the LoS component of the channel from the BS to the m-th LU is given by [15]

$$\bar{\mathbf{h}}_m(\mathbf{t})\triangleq\left[e^{j\frac{2\pi}{\lambda}\mathbf{t}_1^{\mathrm{T}}\mathbf{a}_m},\ldots,e^{j\frac{2\pi}{\lambda}\mathbf{t}_N^{\mathrm{T}}\mathbf{a}_m}\right]^{\mathrm{T}}\in\mathbb{C}^{N\times1}. \tag{1}$$

As in [15], [16], the channel from the BS to the m-th LU follows Rician fading and is given by

$$\mathbf{h}_m(\mathbf{t})=\sqrt{\frac{\kappa_m\beta_m}{\kappa_m+1}}\,\bar{\mathbf{h}}_m(\mathbf{t})+\sqrt{\frac{\beta_m}{\kappa_m+1}}\,\tilde{\mathbf{h}}_m, \tag{2}$$

where β_m is the large-scale fading coefficient, $\kappa_m\geq0$ is the Rician factor, and $\tilde{\mathbf{h}}_m\sim\mathcal{CN}(\mathbf{0},\mathbf{I}_N)$ is the non-LoS (NLoS) component. Similarly, the channel from the BS to Eve is given by

$$\mathbf{g}(\mathbf{t})=\sqrt{\frac{\kappa_{\mathrm{e}}\beta_{\mathrm{e}}}{\kappa_{\mathrm{e}}+1}}\,\bar{\mathbf{g}}(\mathbf{t})+\sqrt{\frac{\beta_{\mathrm{e}}}{\kappa_{\mathrm{e}}+1}}\,\tilde{\mathbf{g}}, \tag{3}$$

where β_{e}, κ_{e}, and $\tilde{\mathbf{g}}\sim\mathcal{CN}(\mathbf{0},\mathbf{I}_N)$ are defined similarly to those in (2), and $\bar{\mathbf{g}}(\mathbf{t})$ is given by (1) with $\mathbf{a}_m$ replaced by $\mathbf{a}_{\mathrm{e}}\triangleq[\cos\theta_{\mathrm{e}}\sin\phi_{\mathrm{e}},\ \sin\theta_{\mathrm{e}}]^{\mathrm{T}}$ for the elevation and azimuth AoDs θ_{e} and ϕ_{e} of Eve. The NLoS components $\tilde{\mathbf{h}}_1,\ldots,\tilde{\mathbf{h}}_M$ and $\tilde{\mathbf{g}}$ are mutually independent. Let $\mathbf{H}(\mathbf{t})\triangleq[\mathbf{h}_1(\mathbf{t}),\ldots,\mathbf{h}_M(\mathbf{t})]\in\mathbb{C}^{N\times M}$ denote the channel matrix from the BS to the LUs, and let $\bar{\mathbf{H}}(\mathbf{t})\triangleq[\bar{\mathbf{h}}_1(\mathbf{t}),\ldots,\bar{\mathbf{h}}_M(\mathbf{t})]$ and $\tilde{\mathbf{H}}\triangleq[\tilde{\mathbf{h}}_1,\ldots,\tilde{\mathbf{h}}_M]$ collect its LoS and NLoS components, respectively. The Rician factors of the LUs are collected in $\mathbf{K}\triangleq\mathrm{diag}(\kappa_1,\ldots,\kappa_M)$.

B. Two-Timescale Transmission Protocol

The statistical CSI is determined by the large-scale fading and the LoS geometry, i.e., by $\{\beta_m,\kappa_m,\theta_m,\phi_m\}_{m\in\mathcal{M}}$ for the LUs and by $\{\beta_{\mathrm{e}},\kappa_{\mathrm{e}},\theta_{\mathrm{e}},\phi_{\mathrm{e}}\}$ for Eve. Hence, it varies much more slowly than the NLoS components $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{g}}$, which

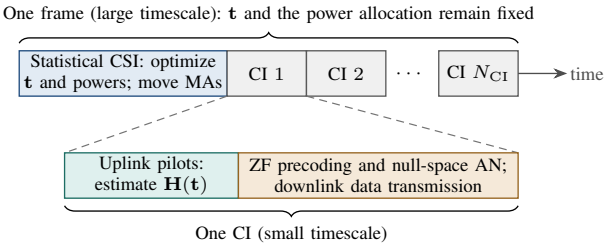


Fig. 2. Two-timescale transmission protocol (CI: coherence interval).

change from one CI to another. Following [12], we assume that the statistical CSI of Eve is known at the BS.[1]

Based on these two timescales, we adopt the transmission protocol in Fig. 2, where the time is divided into frames, and each frame consists of $N_{\mathrm{CI}} \gg 1$ CIs with constant statistical CSI. At the beginning of each frame, the BS optimizes the antenna positions and the long-term power allocation based on the statistical CSI and then moves the MAs to the optimized positions. In each CI of the frame, the BS estimates the channels of the LUs at these positions from uplink pilots by exploiting the channel reciprocity of time-division duplexing and then designs the precoders and the AN. For ease of exposition, the channels of the LUs are assumed to be perfectly estimated, whereas the NLoS component $\tilde{\mathbf{g}}$ of Eve is unknown to the BS. Compared with the designs based on instantaneous CSI, this protocol moves the MAs only once per frame and requires the CSI of the LUs only at the current antenna positions.

C. Signal Model and Ergodic Secrecy Rate

In each CI, the transmit signal of the BS is given by

$$\mathbf{x} = \sum_{m=1}^{M} \mathbf{w}_m s_m + \mathbf{z}, \tag{4}$$

where $s_m \sim \mathcal{CN}(0,1)$ is the confidential symbol for the m-th LU, $\mathbf{w}_m \in \mathbb{C}^{N\times 1}$ is the corresponding precoding vector, and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma})$ is the AN with covariance matrix $\mathbf{\Sigma} \succeq \mathbf{0}$. The symbols $s_1, \ldots, s_M$ and the AN $\mathbf{z}$ are mutually independent, and the transmit power satisfies $\sum_{m=1}^{M} \|\mathbf{w}_m\|^2 + \mathrm{tr}(\mathbf{\Sigma}) \le P_{\mathrm{tot}}$, where P_{tot} is the maximum transmit power of the BS.

The signal received by the m-th LU is given by

$$y_m = \mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\mathbf{x} + n_m, \tag{5}$$

where $n_m \sim \mathcal{CN}(0, \sigma_m^2)$ is the additive white Gaussian noise (AWGN) at the m-th LU. Accordingly, the signal-to-interference-plus-noise ratio (SINR) of the m-th LU is given by

$$\gamma_m = \frac{|\mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\mathbf{w}_m|^2}{\sum_{i=1, i\neq m}^{M} |\mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\mathbf{w}_i|^2 + \mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\mathbf{\Sigma}\mathbf{h}_m(\mathbf{t}) + \sigma_m^2}. \tag{6}$$

Similarly, the signal received by Eve is given by

$$y_{\mathrm{e}} = \mathbf{g}^{\mathrm{H}}(\mathbf{t})\mathbf{x} + n_{\mathrm{e}}, \tag{7}$$

[1]The statistical CSI of Eve can be obtained, e.g., from prior knowledge of its location, by sensing its direction [14], or from its historical signals if it is an untrusted node.

where $n_{\mathrm{e}} \sim \mathcal{CN}(0, \sigma_{\mathrm{e}}^2)$ is the AWGN at Eve. Since the BS does not know the receiver structure of the passive Eve, we consider the worst case where Eve can cancel the signals of the other LUs when intercepting s_m, e.g., via multiuser detection [19]. This worst-case model provides a secrecy guarantee that does not depend on the decoding capability of Eve, and the resulting SINR of Eve for s_m is given by

$$\gamma_{\mathrm{e},m} = \frac{|\mathbf{g}^{\mathrm{H}}(\mathbf{t})\mathbf{w}_m|^2}{\mathbf{g}^{\mathrm{H}}(\mathbf{t})\mathbf{\Sigma}\mathbf{g}(\mathbf{t}) + \sigma_{\mathrm{e}}^2}, \tag{8}$$

which upper-bounds that of an Eve treating the signals of the other LUs as interference.

Since the instantaneous CSI of Eve is unknown, the BS cannot adapt the transmission rate to each realization of $\tilde{\mathbf{g}}$. Instead, the confidential message of each LU is encoded over the N_{CI} CIs of a frame, which experience independent realizations of the NLoS components. With this coding scheme, the following ergodic secrecy rate is achievable for the m-th LU [19], [20]:

$$R_m^{\mathrm{sec}} \triangleq \left[\bar{R}_m - \bar{R}_{\mathrm{e},m}\right]^+, \tag{9}$$

where

$$\bar{R}_m \triangleq \mathbb{E}\{\log_2(1+\gamma_m)\}, \quad \bar{R}_{\mathrm{e},m} \triangleq \mathbb{E}\{\log_2(1+\gamma_{\mathrm{e},m})\} \tag{10}$$

denote the ergodic rate of the m-th LU and the ergodic wiretap rate of Eve for s_m, respectively, with the expectations taken over $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{g}}$ for a given $\mathbf{t}$. The ESSR is then given by $\sum_{m=1}^{M} R_m^{\mathrm{sec}}$.

D. Problem Formulation

Since the precoders and the AN covariance matrix are adapted to the instantaneous CSI of the LUs, they are designed for each realization of $\mathbf{H}(\mathbf{t})$, whereas the antenna positions are designed based on the statistical CSI. Accordingly, the ESSR maximization problem is formulated as

$$\begin{aligned}
(\mathrm{P1}): \max_{\mathbf{t}, \{\mathbf{w}_m\}, \mathbf{\Sigma}} \quad & \sum_{m=1}^{M} R_m^{\mathrm{sec}} && \text{(11a)}\\
\text{s.t.} \quad & \sum_{m=1}^{M} \|\mathbf{w}_m\|^2 + \mathrm{tr}(\mathbf{\Sigma}) \le P_{\mathrm{tot}}, \\
& \mathbf{\Sigma} \succeq \mathbf{0}, \ \forall \mathbf{H}(\mathbf{t}), && \text{(11b)}\\
& \mathbf{t}_n \in \mathcal{C}, \ \forall n \in \mathcal{N}, && \text{(11c)}\\
& \|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min}, \ \forall n, i \in \mathcal{N}, \ n \neq i. && \text{(11d)}
\end{aligned}$$

Problem (P1) is a nonconvex stochastic optimization problem, which is challenging to solve. Specifically, the ergodic rates in (10) have no closed-form expressions, and the precoders and the AN covariance matrix need to be optimized for every realization of $\mathbf{H}(\mathbf{t})$. Moreover, the objective function is nonconcave with respect to $\mathbf{t}$ due to the phases of the LoS components in (1), and the minimum-distance constraints in (11d) are nonconvex. To tackle these challenges, we first adopt ZF precoding and null-space AN with a long-term power allocation and approximate the ergodic rates in closed form in Section III, which yields a deterministic problem in the long-term variables only. Then, an AO algorithm is developed for this deterministic problem in Section IV.

III. Two-Timescale ZF-AN Design and Ergodic Secrecy Rate Analysis

A. ZF Precoding and Null-Space AN

Since the instantaneous CSI of Eve is unavailable, the precoders and the AN cannot be designed to directly reduce the information leakage in each CI. Hence, we adopt ZF precoding and null-space AN [5], [19], which require only the instantaneous CSI of the LUs. Specifically, the precoder of the m-th LU is given by $\mathbf{w}_m = \sqrt{p_m}\,\bar{\mathbf{w}}_m$, where $p_m \geq 0$ is the power allocated to the m-th LU, and

$$\bar{\mathbf{w}}_m \triangleq \frac{\mathbf{H}(\mathbf{t})\big(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m}{\big\|\mathbf{H}(\mathbf{t})\big(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m\big\|} \tag{12}$$

is the normalized ZF precoding vector, which satisfies $\mathbf{h}_i^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m = 0$, $\forall i \neq m$. Moreover, the AN is isotropically distributed in the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$ with the covariance matrix $\mathbf{\Sigma} = q\mathbf{P}^{\perp}(\mathbf{t})$, where $q \geq 0$ is the AN power per dimension, and

$$\mathbf{P}^{\perp}(\mathbf{t}) \triangleq \mathbf{I}_N - \mathbf{H}(\mathbf{t})\big(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{H}^{\mathrm{H}}(\mathbf{t}) \tag{13}$$

is the orthogonal projection onto the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$. Since $\mathbf{H}(\mathbf{t})$ has full column rank with probability one, we have $\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{0}$ and $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$. Hence, the ZF precoders eliminate the inter-user interference, the AN causes no interference to the LUs, and the transmit power in each CI is $\sum_{m=1}^{M} p_m + (N-M)q$.

The powers $\mathbf{p} \triangleq [p_1, \ldots, p_M]^{\mathrm{T}}$ and q constitute the long-term power allocation, which is optimized based on the statistical CSI and remains fixed within a frame, whereas $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are updated in each CI according to $\mathbf{H}(\mathbf{t})$. As a result, the ZF-AN design involves no optimization in each CI. With this design, the SINRs in (6) and (8) reduce to

$$\gamma_m = \frac{p_m|\mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2}{\sigma_m^2} = \frac{p_m}{\sigma_m^2\big[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\big]_{m,m}}, \tag{14}$$

$$\gamma_{\mathrm{e},m} = \frac{p_m|\mathbf{g}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2}{q\,\mathbf{g}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t}) + \sigma_{\mathrm{e}}^2}, \tag{15}$$

where the second equality in (14) follows from $|\mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2 = 1/[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}$ by (12).

The equal power allocation in [15] and the information–AN power splitting with equal power per LU are special cases of $(\mathbf{p}, q)$. With the ZF-AN design, (P1) is restricted to

$$\begin{aligned}
(\mathrm{P2}):\ \max_{\mathbf{t},\mathbf{p},q}\ & \sum_{m=1}^{M} R_m^{\mathrm{sec}}(\mathbf{t}, p_m, q) && \text{(16a)}\\
\text{s.t.}\ & \sum_{m=1}^{M} p_m + (N-M)q \leq P_{\mathrm{tot}}, && \text{(16b)}\\
& p_m \geq 0,\ \forall m \in \mathcal{M}, \quad q \geq 0, && \text{(16c)}\\
& \mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, && \text{(16d)}\\
& \|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min},\ \forall n, i \in \mathcal{N},\ n \neq i, && \text{(16e)}
\end{aligned}$$

where $R_m^{\mathrm{sec}}(\mathbf{t}, p_m, q)$ denotes (9) with the SINRs in (14) and (15), and (16b) and (16c) guarantee (11b) in every CI. Since the ZF-AN design is a particular choice of the precoders and the AN covariance matrix in (P1), (P2) is a restriction of (P1), and its optimal value is a lower bound on that of (P1). However, (P2) is still intractable because the ergodic rates in (16a) have no closed-form expressions. Hence, we approximate the ergodic rates of the LUs and Eve in Sections III-B and III-C, respectively.

B. Ergodic Rate of the LUs

Let $\tilde{\beta}_m \triangleq \beta_m/(\kappa_m + 1)$ denote the power of the NLoS component of the m-th LU, and define $\mathbf{B} \triangleq \mathrm{diag}(\tilde{\beta}_1, \ldots, \tilde{\beta}_M)$ and $\mathbf{L}(\mathbf{t}) \triangleq \bar{\mathbf{H}}(\mathbf{t})\mathbf{K}^{1/2}$. From (2), the channel matrix of the LUs can be written as

$$\mathbf{H}(\mathbf{t}) = \big(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}\big)\mathbf{B}^{1/2}. \tag{17}$$

By (14), the ergodic rate of the m-th LU is determined by the distribution of $[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}$. Since $\log_2(1 + p_m/(\sigma_m^2 z))$ is convex in $z > 0$, applying Jensen's inequality to (14) yields

$$\bar{R}_m \geq \log_2\left(1 + \frac{p_m}{\sigma_m^2\mathbb{E}\big\{[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}\big\}}\right). \tag{18}$$

For Rician fading, the expectation in (18) has no closed-form expression. Following [18], we approximate the Gram matrix $\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t})$ by a central Wishart matrix with N degrees of freedom and the same mean $\mathbf{B}^{1/2}\mathbf{\Phi}(\mathbf{t})\mathbf{B}^{1/2}$, where

$$\mathbf{\Phi}(\mathbf{t}) \triangleq \mathbf{L}^{\mathrm{H}}(\mathbf{t})\mathbf{L}(\mathbf{t}) + N\mathbf{I}_M. \tag{19}$$

Since the inverse of this Wishart matrix has the mean $N\mathbf{B}^{-1/2}\mathbf{\Omega}(\mathbf{t})\mathbf{B}^{-1/2}/(N-M)$ with $\mathbf{\Omega}(\mathbf{t}) \triangleq \mathbf{\Phi}^{-1}(\mathbf{t})$, we substitute this mean into (18) and approximate the ergodic rate of the m-th LU by

$$\bar{R}_m \approx \tilde{R}_m(\mathbf{t}, p_m) \triangleq \log_2\big(1 + \eta_m(\mathbf{t})\,p_m\big), \tag{20}$$

where $\eta_m(\mathbf{t}) \triangleq \zeta_m(N-M)/(N\omega_m(\mathbf{t}))$ is the effective channel gain of the m-th LU, $\zeta_m \triangleq \tilde{\beta}_m/\sigma_m^2$, and $\omega_m(\mathbf{t}) \triangleq [\mathbf{\Omega}(\mathbf{t})]_{m,m}$.

The approximation in (20) is equivalent to that in [15, Sec. IV-A]. For Rayleigh fading, the central Wishart model is exact, and (20) reduces to the well-known lower bound $\log_2(1 + \zeta_m p_m(N-M))$ [21]. For Rician fading, (20) underestimates the ergodic rate in the LoS-dominant regime, as shown in Section V-B. In addition, the antenna positions affect $\tilde{R}_m(\mathbf{t}, p_m)$ only through $\omega_m(\mathbf{t})$. Since $\mathbf{\Phi}(\mathbf{t}) \succeq N\mathbf{I}_M$ and $[\mathbf{\Omega}(\mathbf{t})]_{m,m} \geq 1/[\mathbf{\Phi}(\mathbf{t})]_{m,m} = 1/(N(\kappa_m+1))$ [22], we have

$$\zeta_m(N-M) \leq \eta_m(\mathbf{t}) \leq \zeta_m(N-M)(\kappa_m+1), \tag{21}$$

where the upper bound is achieved if $\bar{\mathbf{h}}_m^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{h}}_i(\mathbf{t}) = 0$ for all $i \neq m$ with $\kappa_i > 0$. Hence, by decorrelating the LoS component of the m-th LU from those of the other LUs, the antenna positions can increase $\eta_m(\mathbf{t})$ by up to a factor of $\kappa_m + 1$.

C. Ergodic Wiretap Rate of Eve

Unlike γ_m in (14), which depends on $\mathbf{H}(\mathbf{t})$ only through $[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}$, the SINR of Eve in (15) depends on the ZF direction $\bar{\mathbf{w}}_m$ and the AN subspace, both of which vary randomly with $\mathbf{H}(\mathbf{t})$. To handle this dependence, we

first characterize the information leakage and the AN power received by Eve for a given $\mathbf{H}(\mathbf{t})$.

Lemma 1: Conditioned on $\mathbf{H}(\mathbf{t})$, the information leakage $X_m \triangleq |\mathbf{g}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2$ and the AN power $Y \triangleq \mathbf{g}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t})$ received by Eve are independent, and their conditional means and variances are given by

$$\begin{aligned}\mathbb{E}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}\big(\kappa_{\mathrm{e}}\bar{X}_m + 1\big),\\ \mathrm{Var}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}^2\big(2\kappa_{\mathrm{e}}\bar{X}_m + 1\big),\end{aligned} \tag{22}$$

$$\begin{aligned}\mathbb{E}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}\big(\kappa_{\mathrm{e}}\bar{Y} + N - M\big),\\ \mathrm{Var}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}^2\big(2\kappa_{\mathrm{e}}\bar{Y} + N - M\big),\end{aligned} \tag{23}$$

where $\tilde{\beta}_{\mathrm{e}} \triangleq \beta_{\mathrm{e}}/(\kappa_{\mathrm{e}}+1)$ is the power of the NLoS component of Eve, and $\bar{X}_m \triangleq |\bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2$ and $\bar{Y} \triangleq \bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$ are the LoS counterparts of X_m and Y, respectively.

Proof: See Appendix A. ∎

Lemma 1 shows that the information leakage and the AN power received by Eve are determined by orthogonal components of the wiretap channel, namely, its component along $\bar{\mathbf{w}}_m$ and that in the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$, respectively. Based on this property, we approximate $\bar{R}_{\mathrm{e},m} = \mathbb{E}\{\log_2(1 + p_m X_m/(qY+\sigma_{\mathrm{e}}^2))\}$ by replacing X_m and Y with their means, as in [18, Lemma 1]. This approximation is accurate when X_m and Y concentrate around their means. By Lemma 1, the conditional relative variances of X_m and Y are small when $\kappa_{\mathrm{e}}\bar{X}_m$ and $\kappa_{\mathrm{e}}\bar{Y} + N - M$ are large, respectively. Since $\bar{X}_m$ and $\bar{Y}$ also vary with $\tilde{\mathbf{H}}$, these conditions alone do not guarantee the concentration. Nevertheless, the approximation is asymptotically exact as $\kappa_{\mathrm{e}} \to \infty$ and $\kappa_m \to \infty$, $\forall m \in \mathcal{M}$, provided that the LoS components of the LUs are linearly independent. Moreover, since the rate is concave in X_m and convex in Y, the approximation is neither an upper nor a lower bound, and its accuracy is evaluated in Section V-B. Using Lemma 1 and taking the expectation over $\tilde{\mathbf{H}}$, we obtain

$$\bar{R}_{\mathrm{e},m} \approx \log_2\left(1 + \frac{p_m\big(\kappa_{\mathrm{e}}\mathbb{E}\{\bar{X}_m\} + 1\big)}{q\big(\kappa_{\mathrm{e}}\mathbb{E}\{\bar{Y}\} + N - M\big) + \bar{\sigma}_{\mathrm{e}}^2}\right), \tag{24}$$

where $\bar{\sigma}_{\mathrm{e}}^2 \triangleq \sigma_{\mathrm{e}}^2/\tilde{\beta}_{\mathrm{e}}$ is the normalized noise power of Eve.

The expectations $\mathbb{E}\{\bar{X}_m\}$ and $\mathbb{E}\{\bar{Y}\}$ in (24), i.e., the average LoS energies of Eve along the ZF direction of the m-th LU and in the AN subspace, still have no closed-form expressions. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are invariant to the scaling of the columns of $\mathbf{H}(\mathbf{t})$, they are determined by $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$ in (17), whose Gram matrix has the mean $\mathbf{\Phi}(\mathbf{t})$. As derived in Appendix B, replacing the inverse of this Gram matrix with $\mathbf{\Omega}(\mathbf{t})$, where the NLoS contribution to the AN subspace is calibrated by the identity $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$, yields $\mathbb{E}\{\bar{X}_m\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{\bar{Y}\} \approx d(\mathbf{t})$ with

$$c_m(\mathbf{t}) \triangleq \frac{\big|[\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})]_m\big|^2 + N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}}{\omega_m(\mathbf{t})}, \tag{25}$$

$$d(\mathbf{t}) \triangleq \frac{(N-M)\big(N - \mathbf{v}^{\mathrm{H}}(\mathbf{t})\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})\big)}{N - M + N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t}))}, \tag{26}$$

where $\mathbf{v}(\mathbf{t}) \triangleq \mathbf{L}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t}) = \mathbf{K}^{1/2}\bar{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$ collects the weighted LoS correlations between Eve and the LUs. Substituting (25) and (26) into (24), we approximate the ergodic wiretap rate of Eve for s_m by

$$\bar{R}_{\mathrm{e},m} \approx \tilde{R}_{\mathrm{e},m}(\mathbf{t}, p_m, q) \triangleq \log_2\left(1 + \frac{\xi_m(\mathbf{t})\, p_m}{q\,\psi(\mathbf{t}) + \bar{\sigma}_{\mathrm{e}}^2}\right), \tag{27}$$

where $\xi_m(\mathbf{t}) \triangleq \kappa_{\mathrm{e}}c_m(\mathbf{t}) + 1$ and $\psi(\mathbf{t}) \triangleq \kappa_{\mathrm{e}}d(\mathbf{t}) + N - M$ represent the normalized information leakage for the m-th LU and the normalized AN power received by Eve, respectively. The approximations $c_m(\mathbf{t})$ and $d(\mathbf{t})$ have the following properties.

Proposition 1: The quantities $c_m(\mathbf{t})$, $d(\mathbf{t})$, $\mathbb{E}\{\bar{X}_m\}$, and $\mathbb{E}\{\bar{Y}\}$ lie in $[0, N]$. Moreover, $c_m(\mathbf{t}) = \mathbb{E}\{\bar{X}_m\}$ and $d(\mathbf{t}) = \mathbb{E}\{\bar{Y}\}$ if $\kappa_m = 0$, $\forall m \in \mathcal{M}$. For linearly independent LoS components of the LUs, $c_m(\mathbf{t}) - \mathbb{E}\{\bar{X}_m\}$ and $d(\mathbf{t}) - \mathbb{E}\{\bar{Y}\}$ vanish as $\mathbf{K} = \kappa\mathbf{K}_0$ with $\kappa \to \infty$ for any diagonal $\mathbf{K}_0 \succ \mathbf{0}$.

Proof: See Appendix B. ∎

Unlike the approximations for ZF precoding in [15], [18], which involve only the diagonal entries of $\mathbf{\Omega}(\mathbf{t})$, (27) captures the LoS correlation between Eve and the LUs via $\mathbf{v}(\mathbf{t})$.

D. Secrecy Mechanisms of the MAs

By (20) and (27), the antenna positions affect the approximate ergodic rates only through $\omega_m(\mathbf{t})$, $c_m(\mathbf{t})$, and $d(\mathbf{t})$. In this subsection, we examine how the antenna positions improve the secrecy performance via these quantities.

1) LoS decorrelation: By (19), (25), and (26), the quantities $\omega_m(\mathbf{t})$, $c_m(\mathbf{t})$, and $d(\mathbf{t})$ are determined by the LoS array correlations among the LUs and between Eve and the LUs, which are given by

$$[\bar{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{H}}(\mathbf{t})]_{i,k} = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}(\mathbf{a}_k - \mathbf{a}_i)}, \quad i,k \in \mathcal{M}, \tag{28}$$

$$[\bar{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})]_{i} = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}(\mathbf{a}_{\mathrm{e}} - \mathbf{a}_i)}, \quad i \in \mathcal{M}, \tag{29}$$

respectively. A weaker LoS correlation between the m-th LU and the other LUs reduces $\omega_m(\mathbf{t})$ and thus increases the effective channel gain $\eta_m(\mathbf{t})$ according to (21). This decorrelation among the LUs is the only mechanism exploited by the rate-oriented MA design [15]. For secure transmission, the MAs offer two additional mechanisms: a smaller $c_m(\mathbf{t})$ reduces the information leakage, and a larger $d(\mathbf{t})$ increases the AN received by Eve. In particular, if $\mathbf{v}(\mathbf{t}) = \mathbf{0}$, i.e., the LoS component of Eve is orthogonal to those of all the LUs, then, for a given $\mathbf{\Phi}(\mathbf{t})$, $c_m(\mathbf{t})$ attains its minimum $N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}/\omega_m(\mathbf{t})$ caused by the NLoS components, and $d(\mathbf{t})$ attains its maximum $N(N-M)/(N-M+N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t})))$. Hence, by decorrelating the LoS component of Eve from those of the LUs, the MAs reduce the information leakage and increase the AN received by Eve at the same time. In contrast, an FPA array can hardly achieve this decorrelation when Eve is located in a direction close to that of an LU. If $\kappa_{\mathrm{e}} = 0$, then $\tilde{R}_{\mathrm{e},m} = \log_2(1+p_m/(q(N-M)+\bar{\sigma}_{\mathrm{e}}^2))$ does not depend on $\mathbf{t}$, since the statistical CSI of Eve carries no spatial information, and only the decorrelation among the LUs remains.

2) Single LU: If $M = 1$, then $\mathbf{\Phi}(\mathbf{t}) = N(\kappa_1 + 1)$ is a scalar, and $\eta_1 = \zeta_1(N-1)(\kappa_1+1)$ does not depend on $\mathbf{t}$.[2] In contrast,

$$c_1(\mathbf{t}) = \frac{\kappa_1 |\bar{\mathbf{h}}_1^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2/N + 1}{\kappa_1 + 1}, \tag{30}$$

$$d(\mathbf{t}) = \frac{(N-1)(\kappa_1+1)}{(N-1)(\kappa_1+1)+1}\Big(N + \frac{1}{\kappa_1+1} - c_1(\mathbf{t})\Big) \tag{31}$$

depend on $\mathbf{t}$ through the LoS correlation $|\bar{\mathbf{h}}_1^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve. Therefore, unlike in the rate-oriented MA design, where the MAs bring no gain for a single user [15, Remark 2], the MAs enhance the secrecy of a single LU by reducing this correlation. For $M = 1$, the identity $\bar{\mathbf{w}}_1\bar{\mathbf{w}}_1^{\mathrm{H}} + \mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{I}_N$ further implies $\bar{X}_1 + \bar{Y} = N$, so that reducing the leakage is equivalent to increasing the AN received by Eve. This identity is nearly preserved by (30) and (31), since $c_1(\mathbf{t}) + d(\mathbf{t}) - N = (c_1(\mathbf{t}) - 1)/((N-1)(\kappa_1+1)+1) \in [-1/4, 1/4]$.

3) High-SNR behavior and AN power: To obtain insight into the AN power, we consider the information–AN power splitting with equal power among the LUs, where $p_m = \alpha P_{\mathrm{tot}}/M$ and $q = (1-\alpha)P_{\mathrm{tot}}/(N-M)$ with the power-splitting factor $\alpha \in (0,1)$. As $P_{\mathrm{tot}} \to \infty$, i.e., in the high signal-to-noise ratio (SNR) regime, the approximate secrecy rate of the m-th LU satisfies

$$\tilde{R}_m - \tilde{R}_{\mathrm{e},m} = \log_2 \frac{\alpha P_{\mathrm{tot}} \eta_m(\mathbf{t})}{M} - \log_2\Big(1 + \frac{\alpha \Theta_m(\mathbf{t})}{1-\alpha}\Big) + o(1), \tag{32}$$

where $o(1)$ vanishes as $P_{\mathrm{tot}} \to \infty$, and $\Theta_m(\mathbf{t}) \triangleq \frac{(N-M)\xi_m(\mathbf{t})}{M\psi(\mathbf{t})}$ is the ratio of the normalized information leakage to the normalized AN power received by Eve. Hence, the high-SNR slope of the approximate ESSR is M regardless of $\mathbf{t}$, and the antenna positions affect only the constant term. The value of α that maximizes the sum of the constant terms in (32) satisfies $\sum_{m=1}^{M}\big[\frac{1}{\alpha} - \frac{1}{1-\alpha} - \frac{\Theta_m - 1}{1+\alpha(\Theta_m-1)}\big] = 0$, and for $\Theta_m = \Theta$, $\forall m \in \mathcal{M}$, it is given in closed form by

$$\alpha^{\star} = \frac{1}{1+\sqrt{\Theta}}. \tag{33}$$

Since the MAs can decrease $c_m(\mathbf{t})$ and increase $d(\mathbf{t})$, they decrease Θ and thus increase $\alpha^{\star}$, so that less power is required for the AN. This equal-power case serves only to provide insight, whereas the long-term power allocation is optimized without this restriction in Section IV-A.

E. Approximate ESSR Maximization Problem

Combining (20) and (27), we approximate the ergodic secrecy rate of the m-th LU by

$$R_m^{\mathrm{sec}} \approx \tilde{R}_m^{\mathrm{sec}}(\mathbf{t}, p_m, q) \triangleq \big[\tilde{R}_m(\mathbf{t}, p_m) - \tilde{R}_{\mathrm{e},m}(\mathbf{t}, p_m, q)\big]^+, \tag{34}$$

which depends only on the statistical CSI. Replacing the ergodic secrecy rates in (P2) with (34) yields the following deterministic problem:

$$(\mathrm{P3}):\ \max_{\mathbf{t},\mathbf{p},q}\ F(\mathbf{t},\mathbf{p},q) \triangleq \sum_{m=1}^{M}\Bigg[\log_2\big(1+\eta_m(\mathbf{t})p_m\big) - \log_2\Big(1 + \frac{\xi_m(\mathbf{t})p_m}{q\psi(\mathbf{t}) + \bar{\sigma}_{\mathrm{e}}^2}\Big)\Bigg] \tag{35a}$$

$$\text{s.t.}\quad \text{(16b)–(16e)},$$

where the operator $[\cdot]^+$ in (34) is dropped without loss of optimality. Specifically, for any feasible $(\mathbf{t}, \mathbf{p}, q)$, setting $p_m = 0$ for every LU whose term in (35a) is negative yields another feasible solution. Since the m-th term depends only on p_m and q for a given $\mathbf{t}$, the objective value of this solution equals the approximate ESSR $\sum_{m=1}^{M}\tilde{R}_m^{\mathrm{sec}}(\mathbf{t}, p_m, q)$ of the original solution. Consequently, (P3) has the same optimal value as its counterpart with $[\cdot]^+$, and $F(\mathbf{t},\mathbf{p},q) \le \sum_{m=1}^{M}\tilde{R}_m^{\mathrm{sec}}(\mathbf{t}, p_m, q)$ holds for any feasible solution.

IV. Proposed Two-Timescale Optimization Algorithm

Problem (P3) is nonconvex, since F is nonconcave with respect to $\mathbf{t}$ and the minimum-distance constraints in (16e) are nonconvex. Moreover, $\mathbf{t}$, $\mathbf{p}$, and q are coupled in the objective function, and $\mathbf{p}$ and q are further coupled by the power constraint in (16b). To tackle (P3), we exploit two properties of its objective function. First, for a given $\mathbf{t}$, the coefficients η_m, ξ_m, and ψ in (35a) are constants, and the optimal $\mathbf{p}$ for a given q can be derived in semi-closed form. Second, for given $\mathbf{p}$ and q, each antenna position affects F only through rank-one terms of $\mathbf{\Phi}(\mathbf{t})$ and $\mathbf{v}(\mathbf{t})$, which allows F to be evaluated at a large number of candidate positions with low complexity. Based on these properties, we alternately optimize the power allocation and the antenna positions in Sections IV-A and IV-B, respectively.

A. Power Allocation Optimization for Given $\mathbf{t}$

For given $\mathbf{t}$, (P3) reduces to

$$(\mathrm{P3.1}):\ \max_{\mathbf{p},q}\ \sum_{m=1}^{M} f_m(p_m; q) \tag{36a}$$

$$\text{s.t.}\quad \text{(16b)},\ \text{(16c)},$$

where

$$f_m(p;q) \triangleq \log_2(1+\eta_m p) - \log_2\big(1+\eta_{\mathrm{e},m}(q)\,p\big), \tag{37}$$

and $\eta_{\mathrm{e},m}(q) \triangleq \xi_m/(q\psi + \bar{\sigma}_{\mathrm{e}}^2)$ denotes the effective channel gain of Eve for s_m, which decreases with q. For brevity, the argument $\mathbf{t}$ is omitted in this subsection. Problem (P3.1) is nonconvex in general, because f_m is not jointly concave in p_m and q. For a given q, however, the optimal $\mathbf{p}$ can be obtained

[2]For $M = 1$, even the exact ergodic rate $\mathbb{E}\{\log_2(1 + p_1\|\mathbf{h}_1(\mathbf{t})\|^2/\sigma_1^2)\}$ does not depend on $\mathbf{t}$, because the distribution of $\|\mathbf{h}_1(\mathbf{t})\|^2$ depends on $\bar{\mathbf{h}}_1(\mathbf{t})$ only through $\|\bar{\mathbf{h}}_1(\mathbf{t})\|^2 = N$.

in semi-closed form. To this end, we first examine the first-order and second-order derivatives of f_m with respect to p, which are given by

$$\frac{\partial f_m}{\partial p} = \frac{\eta_m - \eta_{\mathrm{e},m}}{\ln 2\,(1+\eta_m p)(1+\eta_{\mathrm{e},m} p)}, \tag{38}$$

$$\frac{\partial^2 f_m}{\partial p^2} = \frac{1}{\ln 2}\left[\frac{\eta_{\mathrm{e},m}^2}{(1+\eta_{\mathrm{e},m}p)^2} - \frac{\eta_m^2}{(1+\eta_m p)^2}\right], \tag{39}$$

where the argument q of $\eta_{\mathrm{e},m}$ is omitted. If $\eta_m \le \eta_{\mathrm{e},m}$, then $f_m(p;q) \le 0$ for all $p \ge 0$, and thus $p_m = 0$ is optimal. If $\eta_m > \eta_{\mathrm{e},m}$, then $\eta_m(1+\eta_{\mathrm{e},m}p) - \eta_{\mathrm{e},m}(1+\eta_m p) = \eta_m - \eta_{\mathrm{e},m} > 0$ implies $\frac{\eta_m}{1+\eta_m p} > \frac{\eta_{\mathrm{e},m}}{1+\eta_{\mathrm{e},m}p}$, so that f_m is strictly increasing and strictly concave in $p \ge 0$ by (38) and (39). Let $\mathcal{M}_{\mathrm{a}}(q) \triangleq \{m \in \mathcal{M} \,|\, \eta_m > \eta_{\mathrm{e},m}(q)\}$ denote the set of LUs that can achieve positive secrecy rates, and let $P_{\mathrm{s}}(q) \triangleq P_{\mathrm{tot}} - (N-M)q$ denote the power available for the confidential signals. Then, the optimal $\mathbf{p}$ for a given q is given by the following proposition.

Proposition 2 (Secrecy Water-Filling): For a given q with $P_{\mathrm{s}}(q) > 0$ and $\mathcal{M}_{\mathrm{a}}(q) \neq \emptyset$, the optimal power allocation is $p_m^\star = 0$ for $m \notin \mathcal{M}_{\mathrm{a}}(q)$ and

$$p_m^\star(\nu) = \left[\frac{2(\rho_m - 1)}{\sqrt{(\eta_m - \eta_{\mathrm{e},m})^2 + 4\eta_m\eta_{\mathrm{e},m}\rho_m} + \eta_m + \eta_{\mathrm{e},m}}\right]^+ \tag{40}$$

for $m \in \mathcal{M}_{\mathrm{a}}(q)$, where $\rho_m = \rho_m(\nu) \triangleq (\eta_m - \eta_{\mathrm{e},m})/(\nu \ln 2)$, and the water level $\nu > 0$ is the unique solution of $\sum_{m\in\mathcal{M}_{\mathrm{a}}(q)} p_m^\star(\nu) = P_{\mathrm{s}}(q)$.

Proof: Since the objective function is strictly concave and increasing in $\{p_m\}_{m\in\mathcal{M}_{\mathrm{a}}(q)}$ and Slater's condition holds, the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality, and the power constraint is active at the optimal solution [23]. Let $\nu > 0$ denote the Lagrange multiplier associated with the power constraint. Then, the KKT conditions for $m \in \mathcal{M}_{\mathrm{a}}(q)$ reduce to

$$\left.\frac{\partial f_m}{\partial p}\right|_{p=p_m^\star} = \nu \;\text{ if } p_m^\star > 0, \qquad \left.\frac{\partial f_m}{\partial p}\right|_{p=0} \le \nu \;\text{ if } p_m^\star = 0. \tag{41}$$

By (38), the first condition is equivalent to the quadratic equation $(1+\eta_m p)(1+\eta_{\mathrm{e},m}p) = \rho_m(\nu)$ in p, whose positive root is given by (40) in a form that avoids numerical cancellation. The second condition is equivalent to $\rho_m(\nu) \le 1$, for which (40) also yields $p_m^\star = 0$. Finally, as ν increases from 0 to $\max_{m\in\mathcal{M}_{\mathrm{a}}(q)}(\eta_m - \eta_{\mathrm{e},m})/\ln 2$, $\sum_{m\in\mathcal{M}_{\mathrm{a}}(q)} p_m^\star(\nu)$ decreases continuously and strictly from $+\infty$ to 0. Hence, the water level that satisfies the power constraint with equality is unique. ∎

Since $p_m^\star(\nu)$ is nonincreasing in ν, the water level in Proposition 2 can be found by bisection. As $\eta_{\mathrm{e},m} \to 0$, (40) reduces to the classic water-filling solution $[1/(\nu\ln 2) - 1/\eta_m]^+$. Hence, (40) can be regarded as a secrecy water-filling solution, which allocates zero power to the m-th LU if $\eta_m \le \eta_{\mathrm{e},m}(q)$, i.e., if its effective channel gain does not exceed that of Eve for s_m.

Let $F^\star(q) \triangleq \sum_{m\in\mathcal{M}} f_m(p_m^\star; q)$ denote the optimal value of (P3.1) for a given q, with $F^\star(q) = 0$ if $\mathcal{M}_{\mathrm{a}}(q) = \emptyset$ or $P_{\mathrm{s}}(q) = 0$. The AN power per dimension is then optimized as

$$q^\star = \arg \max_{0\le q\le q_{\max}} \; F^\star(q), \tag{42}$$

where $q_{\max} \triangleq P_{\mathrm{tot}}/(N-M)$. Increasing q reduces the effective channel gains of Eve at the cost of less power for the confidential signals. Since $F^\star(q)$ is not necessarily unimodal, (42) is solved by a coarse grid search followed by a finer one around the best grid point, as summarized in Algorithm 1.

Algorithm 1 Power Allocation for Given $\mathbf{t}$

Input: $\{\eta_m, \xi_m\}_{m\in\mathcal{M}}$, ψ, $\bar{\sigma}_{\mathrm{e}}^2$, P_{tot}, and the grid size Q.
Output: $q^\star$ and $\mathbf{p}^\star$.
1: **for** each q in a uniform grid of Q points over $[0, q_{\max}]$ **do**
2: Obtain $\mathbf{p}^\star$ by Proposition 2, where ν is found by bisection.
3: Evaluate $F^\star(q)$.
4: **end for**
5: Repeat Steps 1–4 over a uniform grid of Q points between the two neighbors of the best grid point.
6: **return** the best q as $q^\star$ and its optimal power allocation as $\mathbf{p}^\star$.

B. Antenna Position Optimization for Given $\mathbf{p}$ *and* q

For given $\mathbf{p}$ and q, (P3) reduces to

$$(\mathrm{P3.2}): \max_{\mathbf{t}} \; F(\mathbf{t}) \tag{43a}$$

$$\text{s.t.} \;\; \mathbf{t}_n \in \mathcal{C}, \; \forall n \in \mathcal{N}, \tag{43b}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min}, \; \forall n, i \in \mathcal{N}, \; n \neq i, \tag{43c}$$

where $F(\mathbf{t}) \triangleq F(\mathbf{t}, \mathbf{p}, q)$. Problem (P3.2) is challenging for two reasons. First, by (28) and (29), $F(\mathbf{t})$ depends on each $\mathbf{t}_n$ through the phases $\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}(\mathbf{a}_k - \mathbf{a}_i)$, $i, k \in \mathcal{M}\cup\{\mathrm{e}\}$. Since each phase varies with the period $\lambda/\|\mathbf{a}_k - \mathbf{a}_i\|$ along $\mathbf{a}_k - \mathbf{a}_i$, $F(\mathbf{t})$ oscillates over distances on the order of a wavelength and has many local maxima within $\mathcal{C}$. Second, the minimum-distance constraints are nonconvex. To reduce the problem dimension, we optimize the antenna positions in an element-wise manner similar to [15], i.e., $\mathbf{t}_n$ is updated with the other positions fixed, for $n = 1, \ldots, N$ in turn. Let $F(\mathbf{t}_n)$ denote F as a function of $\mathbf{t}_n$, with the other positions fixed at their latest values. The subproblem for the n-th MA is given by

$$(\mathrm{P3.2}.n): \max_{\mathbf{t}_n} \; F(\mathbf{t}_n) \tag{44a}$$

$$\text{s.t.} \;\; \mathbf{t}_n \in \mathcal{C}, \tag{44b}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min}, \; \forall i \in \mathcal{N} \setminus \{n\}. \tag{44c}$$

Although (P3.2.n) has only two real variables, $F(\mathbf{t}_n)$ is multimodal. Hence, a local method, such as the MM method in [15], tends to converge to a local maximum close to the current position, which may trap the AO iterations at an inferior solution (see Section V-C). To avoid this, we solve (P3.2.n) in two stages: Stage 1 selects the best candidate position by a grid search over $\mathcal{C}$, and Stage 2 refines this position by the MM method. The cost of the grid search is reduced by the following rank-one structure.

1) Rank-one structure: Let $\boldsymbol{\ell}_n(\mathbf{t}_n) \in \mathbb{C}^{M\times 1}$ denote the conjugate transpose of the n-th row of $\mathbf{L}(\mathbf{t})$, i.e.,

$$\boldsymbol{\ell}_n(\mathbf{t}_n) = \mathbf{K}^{1/2}\left[e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_1}, \ldots, e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_M}\right]^{\mathrm{T}}. \tag{45}$$

Since $\mathbf{L}^{\mathrm{H}}\mathbf{L} = \sum_{i=1}^{N} \boldsymbol{\ell}_i \boldsymbol{\ell}_i^{\mathrm{H}}$ and $\mathbf{L}^{\mathrm{H}}\bar{\mathbf{g}} = \sum_{i=1}^{N} \boldsymbol{\ell}_i e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}$, we have

$$\boldsymbol{\Phi}(\mathbf{t}) = \boldsymbol{\Gamma}_n + \boldsymbol{\ell}_n(\mathbf{t}_n)\boldsymbol{\ell}_n^{\mathrm{H}}(\mathbf{t}_n), \tag{46}$$

$$\mathbf{v}(\mathbf{t}) = \mathbf{r}_n + \boldsymbol{\ell}_n(\mathbf{t}_n)e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}, \tag{47}$$

where $\boldsymbol{\Gamma}_n \triangleq \sum_{i\neq n} \boldsymbol{\ell}_i(\mathbf{t}_i)\boldsymbol{\ell}_i^{\mathrm{H}}(\mathbf{t}_i) + N\mathbf{I}_M$ and $\mathbf{r}_n \triangleq \sum_{i\neq n} \boldsymbol{\ell}_i(\mathbf{t}_i)e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}$ do not depend on $\mathbf{t}_n$. By the Sherman–Morrison formula [22],

$$\boldsymbol{\Omega}(\mathbf{t}) = \boldsymbol{\Gamma}_n^{-1} - \frac{\boldsymbol{\Gamma}_n^{-1}\boldsymbol{\ell}_n(\mathbf{t}_n)\boldsymbol{\ell}_n^{\mathrm{H}}(\mathbf{t}_n)\boldsymbol{\Gamma}_n^{-1}}{1 + \boldsymbol{\ell}_n^{\mathrm{H}}(\mathbf{t}_n)\boldsymbol{\Gamma}_n^{-1}\boldsymbol{\ell}_n(\mathbf{t}_n)}. \tag{48}$$

Hence, given $\boldsymbol{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, evaluating ω_m, c_m, d, and thus $F(\mathbf{t}_n)$ at a candidate position requires only $\mathcal{O}(M^2)$ operations, compared with $\mathcal{O}(NM^2+M^3)$ operations for computing them from scratch.

2) Stage 1 (grid search): Let $\mathcal{G} \subset \mathcal{C}$ denote a uniform grid over $\mathcal{C}$, $\mathbf{t}_n^{\mathrm{cur}}$ the current position of the n-th MA, and $\mathcal{G}_n \triangleq \{\mathbf{t}_n \in \mathcal{G} \,|\, \|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min},\ \forall i \neq n\}$ the set of grid points that are feasible for (P3.2.n). Using (48), we evaluate $F(\mathbf{t}_n)$ at all the candidates in $\mathcal{G}_n \cup \{\mathbf{t}_n^{\mathrm{cur}}\}$ and obtain

$$\hat{\mathbf{t}}_n = \arg \max_{\mathbf{t}_n \in \mathcal{G}_n \cup \{\mathbf{t}_n^{\mathrm{cur}}\}} F(\mathbf{t}_n), \tag{49}$$

which requires $\mathcal{O}(|\mathcal{G}|M^2)$ operations. Since the current position is included in the candidates, $F(\hat{\mathbf{t}}_n) \geq F(\mathbf{t}_n^{\mathrm{cur}})$. The grid spacing is set to a fraction of the shortest period $\lambda/\max_{i,k\in\mathcal{M}\cup\{\mathrm{e}\}} \|\mathbf{a}_k - \mathbf{a}_i\|$ of the above phases, which is at least $\lambda/2$ because $\|\mathbf{a}_i\| \leq 1$. Since the selected grid point is further refined in Stage 2, a moderate spacing suffices, as confirmed numerically in Section V-A.

3) Stage 2 (MM refinement): Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, we refine the position by the MM method [24], where $\mathbf{t}_n^{(k)}$ denotes the position of the n-th MA in the k-th MM iteration. To construct a surrogate function, we derive the gradient of $F(\mathbf{t}_n)$. Let $u_n \in \{x_n, y_n\}$ denote the coordinate of $\mathbf{t}_n$ along the axis $u \in \{x, y\}$, and define $\boldsymbol{\Lambda}_u \triangleq \mathrm{diag}(a_{1,u}, \ldots, a_{M,u})$, where $a_{m,x} \triangleq \cos\theta_m \sin\phi_m$ and $a_{m,y} \triangleq \sin\theta_m$ are the entries of $\mathbf{a}_m$, and $a_{\mathrm{e},u}$ is the corresponding entry of $\mathbf{a}_{\mathrm{e}}$. From (45)–(47), the derivatives of $\boldsymbol{\ell}_n$, $\mathbf{v}$, and $\boldsymbol{\Omega}$ with respect to u_n are given by

$$\dot{\boldsymbol{\ell}}_n \triangleq \frac{\partial \boldsymbol{\ell}_n}{\partial u_n} = -j\frac{2\pi}{\lambda}\boldsymbol{\Lambda}_u\boldsymbol{\ell}_n, \tag{50}$$

$$\dot{\mathbf{v}} \triangleq \frac{\partial \mathbf{v}}{\partial u_n} = j\frac{2\pi}{\lambda}\big(a_{\mathrm{e},u}\mathbf{I}_M - \boldsymbol{\Lambda}_u\big)\boldsymbol{\ell}_n e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}, \tag{51}$$

$$\dot{\boldsymbol{\Omega}} \triangleq \frac{\partial \boldsymbol{\Omega}}{\partial u_n} = -\boldsymbol{\Omega}\big(\dot{\boldsymbol{\ell}}_n\boldsymbol{\ell}_n^{\mathrm{H}} + \boldsymbol{\ell}_n\dot{\boldsymbol{\ell}}_n^{\mathrm{H}}\big)\boldsymbol{\Omega}, \tag{52}$$

where (52) follows from $\partial\boldsymbol{\Phi}^{-1} = -\boldsymbol{\Phi}^{-1}(\partial\boldsymbol{\Phi})\boldsymbol{\Phi}^{-1}$. Using (50)–(52), the derivatives $\dot{\omega}_m \triangleq \partial\omega_m/\partial u_n$, $\dot{c}_m \triangleq \partial c_m/\partial u_n$, and $\dot{d} \triangleq \partial d/\partial u_n$ are derived in closed form in Appendix C. Then, the derivatives of the effective channel gains η_m and $\eta_{\mathrm{e},m} = \xi_m/(q\psi + \bar{\sigma}_{\mathrm{e}}^2)$ are given by $\dot{\eta}_m \triangleq \partial\eta_m/\partial u_n = -\eta_m\dot{\omega}_m/\omega_m$ and $\dot{\eta}_{\mathrm{e},m} \triangleq \partial\eta_{\mathrm{e},m}/\partial u_n = \kappa_{\mathrm{e}}\eta_{\mathrm{e},m}\big(\dot{c}_m/\xi_m - q\dot{d}/(q\psi+\bar{\sigma}_{\mathrm{e}}^2)\big)$, respectively. Since $F(\mathbf{t}_n) = \sum_{m=1}^{M}\big[\log_2(1+\eta_m p_m) - \log_2(1+\eta_{\mathrm{e},m}p_m)\big]$, the entries of the gradient $\nabla_{\mathbf{t}_n}F = [\partial F/\partial x_n,\ \partial F/\partial y_n]^{\mathrm{T}}$ are given by

$$\frac{\partial F}{\partial u_n} = \frac{1}{\ln 2}\sum_{m=1}^{M}\left[\frac{p_m\dot{\eta}_m}{1+\eta_m p_m} - \frac{p_m\dot{\eta}_{\mathrm{e},m}}{1+\eta_{\mathrm{e},m}p_m}\right]. \tag{53}$$

Since $F(\mathbf{t}_n)$ is twice continuously differentiable and $\mathcal{C}$ is compact, $\nabla_{\mathbf{t}_n}F$ is Lipschitz continuous over $\mathcal{C}$. By the descent lemma [25, Lemma 5.7], for any δ_n no smaller than this Lipschitz constant, the following quadratic function is a lower bound of $F(\mathbf{t}_n)$ that is tight at $\mathbf{t}_n^{(k)}$:

$$F(\mathbf{t}_n) \geq \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \triangleq F(\mathbf{t}_n^{(k)}) + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})^{\mathrm{T}}\big(\mathbf{t}_n - \mathbf{t}_n^{(k)}\big) - \frac{\delta_n}{2}\big\|\mathbf{t}_n - \mathbf{t}_n^{(k)}\big\|^2. \tag{54}$$

An upper bound on the Lipschitz constant could be obtained by bounding the Hessian of $F(\mathbf{t}_n)$ over $\mathcal{C}$. For the composite function $F(\mathbf{t}_n)$, however, such a global bound is difficult to derive and generally loose, which would lead to small MM steps. Hence, δ_n is determined by backtracking [25]: starting from $\max\{\delta_0, \delta_n^{\mathrm{pre}}/2\}$, where $\delta_0 > 0$ is a constant and δ_n^{pre} is the previously accepted value, δ_n is doubled until the update below satisfies $F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)})$, which holds once δ_n reaches the Lipschitz constant. In this way, δ_n adapts to the local curvature of $F(\mathbf{t}_n)$, while the monotonicity of the MM updates is preserved.

For the minimum-distance constraints, $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ is convex with respect to $\mathbf{t}_n$ and is therefore lower-bounded by its first-order Taylor expansion at $\mathbf{t}_n^{(k)}$ [15], i.e.,

$$\|\mathbf{t}_n - \mathbf{t}_i\|^2 \geq \big(\mathbf{t}_n^{(k)} - \mathbf{t}_i\big)^{\mathrm{T}}\big(2\mathbf{t}_n - \mathbf{t}_n^{(k)} - \mathbf{t}_i\big). \tag{55}$$

Replacing $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ in the minimum-distance constraints with the right-hand side of (55) yields the linear constraints

$$\big(\mathbf{t}_n^{(k)} - \mathbf{t}_i\big)^{\mathrm{T}}\big(2\mathbf{t}_n - \mathbf{t}_n^{(k)} - \mathbf{t}_i\big) \geq D_{\min}^2, \quad \forall i \neq n, \tag{56}$$

whose feasible set is a convex subset of that of the minimum-distance constraints with respect to $\mathbf{t}_n$. Combining (54) and (56), we update $\mathbf{t}_n$ by solving

$$\mathbf{t}_n^{(k+1)} = \arg\max_{\mathbf{t}_n}\ \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \quad \text{s.t. } \mathbf{t}_n \in \mathcal{C},\ (56). \tag{57}$$

Completing the square shows that (57) is equivalent to projecting $\mathbf{t}_n^{(k)} + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})/\delta_n$ onto the convex polygon defined by $\mathcal{C}$ and (56). The solution is this point itself if it is feasible and otherwise lies on an edge or at a vertex of the polygon, so that enumerating the projections onto the $N+3$ constraint lines and their pairwise intersections yields the exact solution. Since $\mathbf{t}_n^{(k)}$ is feasible for (57), the update satisfies

$$F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)}) \geq \hat{F}_n(\mathbf{t}_n^{(k)}|\mathbf{t}_n^{(k)}) = F(\mathbf{t}_n^{(k)}), \tag{58}$$

and $\mathbf{t}_n^{(k+1)}$ is feasible for (P3.2.n) due to (55). The MM iterations are repeated until the increase of F normalized by $\max\{1, |F|\}$ is below ϵ_{in} or N_{MM} iterations are reached.

Algorithm 2 Two-Timescale AO Algorithm for (P3)

Input: Statistical CSI, P_{tot}, $\mathcal{C}$, $D_{\min}$, Q, $\mathcal{G}$, δ_0, ϵ_{in}, N_{MM}, ϵ, N_{AO}.
Output: $\mathbf{t}^\star$, $\mathbf{p}^\star$, and $q^\star$.

```
Initialize feasible positions t and obtain (p, q) by Algorithm 1.
repeat
    for n = 1 to N do
        Stage 1: Compute Γ_n^{-1} and r_n, and obtain t̂_n by (49).
        Stage 2: Starting from t_n^{(0)} = t̂_n, update t_n by (57)
    with backtracking on δ_n until the increase of F normalized by
    max{1, |F|} is below ε_in or N_MM iterations are reached.
    end for
    Obtain (p', q') by Algorithm 1 for the updated t, and set
    (p, q) ← (p', q') if F(t, p', q') ≥ F(t, p, q).
until the increase of F normalized by max{1, |F|} is below ε
    or N_AO iterations are reached
```

C. Overall Algorithm, Convergence, and Complexity

Algorithm 2 summarizes the overall AO algorithm for (P3), whose convergence, initialization, and complexity are discussed below.

Convergence: In each iteration of Algorithm 2, the grid search in (49) does not decrease the objective value of (P3) because the current position is among the candidates, and neither do the MM updates in (57) due to (58). The power allocation is likewise accepted only if the objective value does not decrease, so the objective value is nondecreasing over the iterations. It is also upper-bounded by $\sum_{m=1}^{M} \log_2(1+\zeta_m(N-M)(\kappa_m+1)P_{\mathrm{tot}})$ according to (21) and $p_m \le P_{\mathrm{tot}}$, and it is therefore guaranteed to converge. This guarantee concerns the objective value only; it does not imply that the iterates converge to a stationary point or a globally optimal solution of (P3).

Initialization: Since (P3) is nonconvex, the solution obtained by Algorithm 2 may depend on the initial positions. This dependence is weaker than for purely local updates, because the grid search in Stage 1 allows each MA to move to any feasible grid point in $\mathcal{C}$ in every iteration. In addition, Algorithm 2 can be run from several initializations, e.g., the positions of the FPA array and random feasible positions, and the best solution is retained.

Complexity: In each AO iteration, forming $\mathbf{\Phi}$ and $\mathbf{\Omega}$ requires $\mathcal{O}(NM^2+M^3)$ operations. For each MA, $\mathbf{\Gamma}_n^{-1}$ is obtained from $\mathbf{\Omega}$ by a rank-one update. Stage 1 requires $\mathcal{O}(|\mathcal{G}|(M^2+N))$ operations, including the feasibility check of the grid points. In Stage 2, the gradient in (53) and each evaluation of F require $\mathcal{O}(M^2)$ operations based on (48), and solving (57) by enumeration requires $\mathcal{O}(N^3)$ operations. Algorithm 1 requires $\mathcal{O}(QN_{\mathrm{b}}M)$ operations, where N_{b} is the number of bisection iterations. The complexity of Algorithm 2 per initialization is therefore $\mathcal{O}\big(N_{\mathrm{AO}}\big(NM^2+M^3+N|\mathcal{G}|(M^2+N)+NN_{\mathrm{MM}}N_{\mathrm{bt}}(M^2+N^3)+QN_{\mathrm{b}}M\big)\big)$, where N_{bt} is the average number of backtracking trials. In contrast to stochastic optimization [17], this complexity does not depend on the number of channel realizations. Since Algorithm 2 depends only on the slowly varying statistical CSI, it is executed only once per frame, whereas the ZF precoding and AN generation in each CI require $\mathcal{O}(NM^2+M^3)$ operations.

V. Simulation Results

A. Simulation Setup and Benchmarks

Unless otherwise specified, we set $N=8$, $M=3$, $A=4\lambda$, $D_{\min}=\lambda/2$, $\kappa_m=\kappa_{\mathrm{e}}=10$ dB, $\beta_m=\beta_{\mathrm{e}}=1$, and $\sigma_m^2=\sigma_{\mathrm{e}}^2=\sigma^2$, so that $P_{\mathrm{tot}}/\sigma^2$ denotes the transmit SNR, which is set to 20 dB by default. The elevation and azimuth AoDs of the LUs are independently and uniformly distributed within $[-\pi/3,\pi/3]$. Similar to [14], Eve is located in a direction close to that of LU 1, which represents the challenging case discussed in Section I. Its AoDs are $\theta_{\mathrm{e}}=\theta_1+\Delta\cos\tilde{\phi}$ and $\phi_{\mathrm{e}}=\phi_1+\Delta\sin\tilde{\phi}$, so that they deviate from those of LU 1 by the AoD offset Δ in the direction specified by $\tilde{\phi}$, which is uniformly distributed over $[0,2\pi)$. Note that Δ is an offset in the elevation–azimuth coordinates rather than the physical angle between the two directions. By default, $\Delta=0.1$ rad, for which the normalized LoS correlation $|\bar{\mathbf{h}}_1^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|/N$ between Eve and LU 1 exceeds 0.94 for the FPA; Eve thus lies well within the main lobe of the beam steered toward LU 1.

For Algorithm 2, the grid $\mathcal{G}$ in Stage 1 has a spacing of $\lambda/10$, which is at most one fifth of the shortest phase period in Section IV-B and yields $|\mathcal{G}|=1681$ for $A=4\lambda$. Halving this spacing changes the average ESSR of the proposed design by only 0.01% over 20 AoD realizations. The remaining parameters are $\delta_0=1$, $\epsilon_{\mathrm{in}}=10^{-6}$, $N_{\mathrm{MM}}=20$, $\epsilon=10^{-4}$, $N_{\mathrm{AO}}=50$, $Q=41$, and $N_{\mathrm{b}}=60$. Algorithm 2 is executed from a $2\times(N/2)$ uniform planar array (UPA) with $\lambda/2$ spacing and two sets of random feasible positions, and the best solution is selected. All the results are averaged over 100 random AoD realizations, which are shared by all the schemes so that the comparisons are paired; the standard error of the average ESSR of the proposed design at 20 dB is 0.55% of its value. Since the closed-form approximations are used only for optimization, the ergodic rates of each design are evaluated with the exact SINRs in (14) and (15) over 5×10^3 channel realizations (10^4 in Fig. 3).

We compare the proposed design with four benchmarks: 1) **Rate-oriented MA**: The antenna positions maximize the approximate ergodic sum rate $\sum_{m=1}^{M}\log_2(1+\eta_m(\mathbf{t})P_{\mathrm{tot}}/M)$ of the LUs with equal power and ignore Eve [15]. For a fair comparison, this problem is tackled by the position update in Section IV-B with the same initializations. This benchmark isolates the gain of the secrecy-aware positioning enabled by (27). 2) **AS**: Similar to [14], the BS selects N of $2N$ candidate antennas, which form a UPA with 4 rows, $N/2$ columns, and $\lambda/2$ spacing. The subsets are ranked by the approximate ESSR in (P3) with the AN power searched over 21 grid points. For $N\le10$, all the $\binom{2N}{N}$ subsets, e.g., 12870 for $N=8$, are ranked exhaustively, whereas a swap search is adopted for larger N (see Section V-E). The power allocation of the selected subset is refined by Algorithm 1. AS requires no antenna movement but twice as many antennas and an RF switching network. 3) **Sparse FPA**: A $2\times(N/2)$ UPA spans $\mathcal{C}$ with spacings $2A/(N-2)$ and A along the x- and y-axes, respectively. This benchmark has the same aperture as the MAs and thus separates the gain of the optimized positions from that of the large aperture. 4) **FPA**: The BS adopts the $2\times(N/2)$ UPA with $\lambda/2$ spacing. All the schemes use the

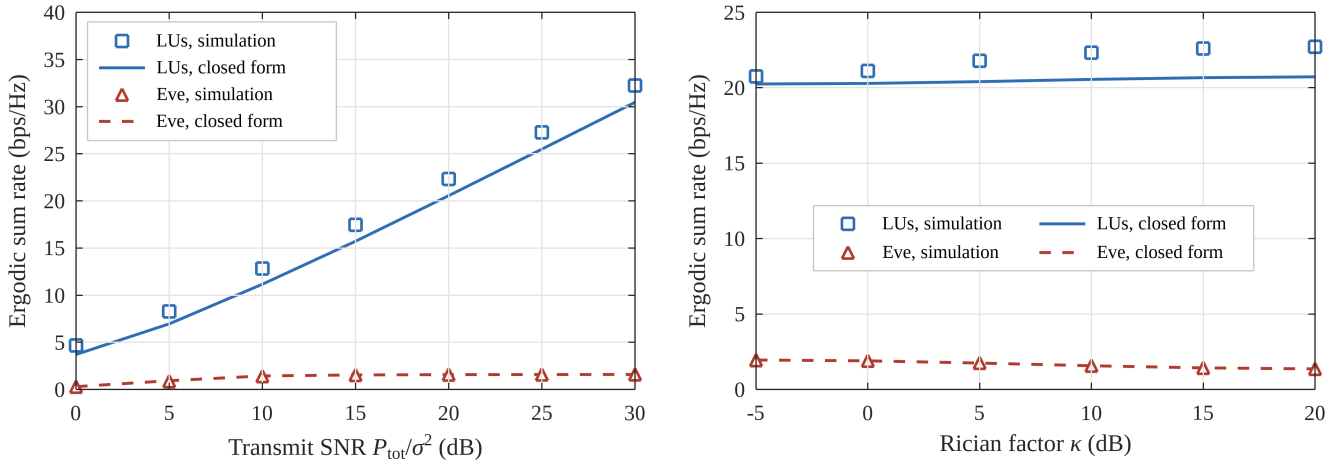


Fig. 3. Ergodic sum rates of the LUs and Eve for the proposed design by Monte Carlo simulations (markers) and by (20) and (27) (lines) versus (a) the transmit SNR P_{tot}/σ^2 and (b) the Rician factor $\kappa = \kappa_m = \kappa_e$.

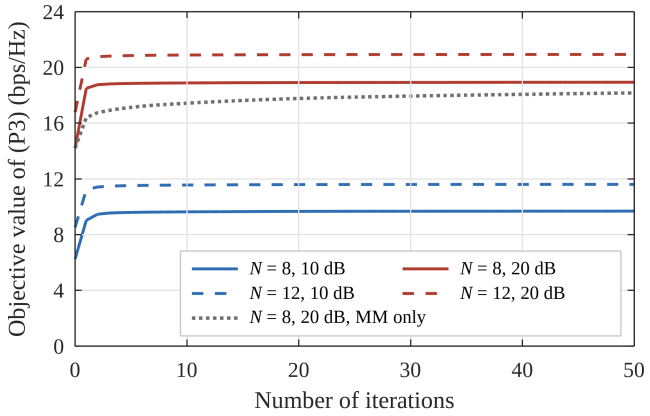


Fig. 4. Convergence behavior of Algorithm 2 for different numbers of MAs N and transmit SNRs P_{tot}/σ^2, averaged over 100 AoD realizations (UPA initialization). "MM only": without the grid search in Stage 1.

power allocation in Algorithm 1, so that their performance differences result only from the antenna positions.

B. Accuracy of the Closed-Form Approximations

Fig. 3 evaluates the accuracy of (20) and (27) at the designs obtained by Algorithm 2. The approximate wiretap rate in (27) agrees with the Monte Carlo results within maximum absolute errors of 0.052 and 0.013 bps/Hz over the SNRs in Fig. 3(a) and the Rician factors in Fig. 3(b), respectively, which indicates that it captures the LoS correlation between Eve and the LUs. By contrast, (20) underestimates the ergodic sum rate of the LUs, and the gap grows with the Rician factor. This underestimation stems from Jensen's inequality and from the central Wishart approximation in Section III-B, which is exact for Rayleigh fading but overestimates $\mathbb{E}\{[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}\}$ by a factor approaching $N/(N-M)$ as the LoS components dominate. As a result, the approximate ESSR in (34) is lower than the actual ESSR in all the tested cases, e.g., 18.98 versus 20.75 bps/Hz at 20 dB. Since this gap is similar for all the schemes, namely between 1.52 and 1.76 bps/Hz at 20 dB, and smaller than the gains of the proposed design, the approximation remains a suitable design objective.

C. Convergence Behavior of Algorithm 2

Fig. 4 shows the objective value of (P3) versus the number of iterations of Algorithm 2, which is run without the stopping criterion. In all the settings, the objective value increases monotonically and converges within about 15 iterations. This fast convergence results from the grid search in Stage 1, which moves each MA directly to the best feasible grid point instead of letting it traverse the movable region through local updates. With the MM updates only, the objective value increases much more slowly and remains lower, because the MAs are trapped in the local maxima near their initial positions.

D. ESSR and AN Power Versus Transmit SNR

Fig. 5(a) shows that the proposed design achieves the highest ESSR over the entire SNR range, e.g., 14.8% and 31.2% higher than those of the rate-oriented MA and the FPA at 20 dB, respectively. Although the rate-oriented MA exploits the same movable region, its gain over the FPA is less than half of that of the proposed design, because its positions only decorrelate the LUs, whereas the proposed positions also reduce the information leakage and increase the AN received by Eve (see Section III-D). AS achieves an even smaller gain, since its $2N$ candidates form a compact array whose spatial resolution is insufficient to decorrelate the LoS components of Eve and LU 1. The sparse FPA has the same aperture as the MAs, yet it also achieves less than half of the gain of the proposed design over the FPA, because its fixed positions cannot adapt the LoS correlations to the AoDs of the LUs and Eve. The secrecy gain of the MAs therefore stems mainly from the optimized positions rather than from

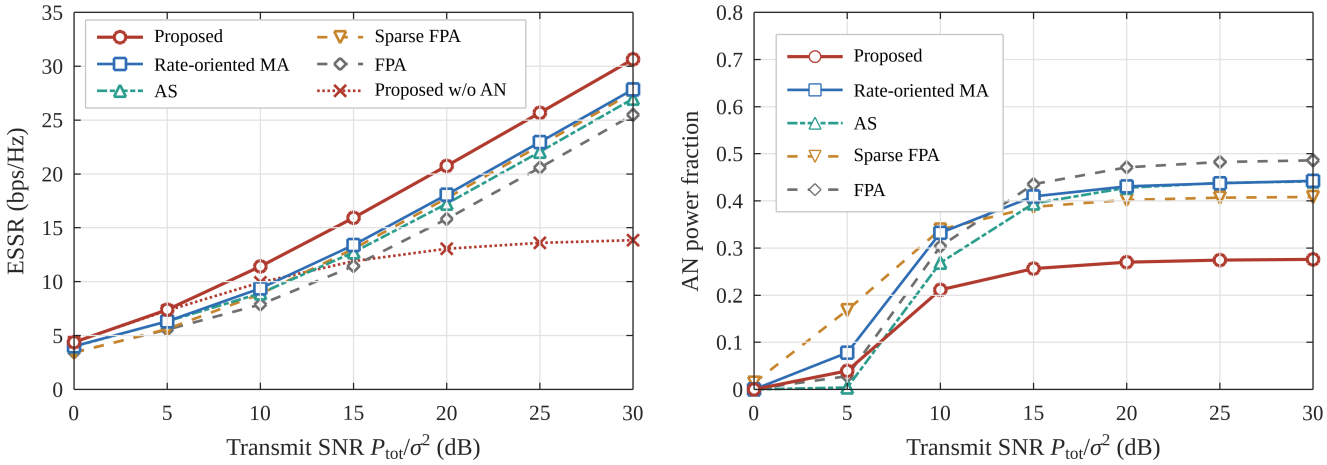


Fig. 5. (a) ESSR and (b) optimized AN power fraction $(N-M)q/P_{\text{tot}}$ versus the transmit SNR P_{tot}/σ^2.

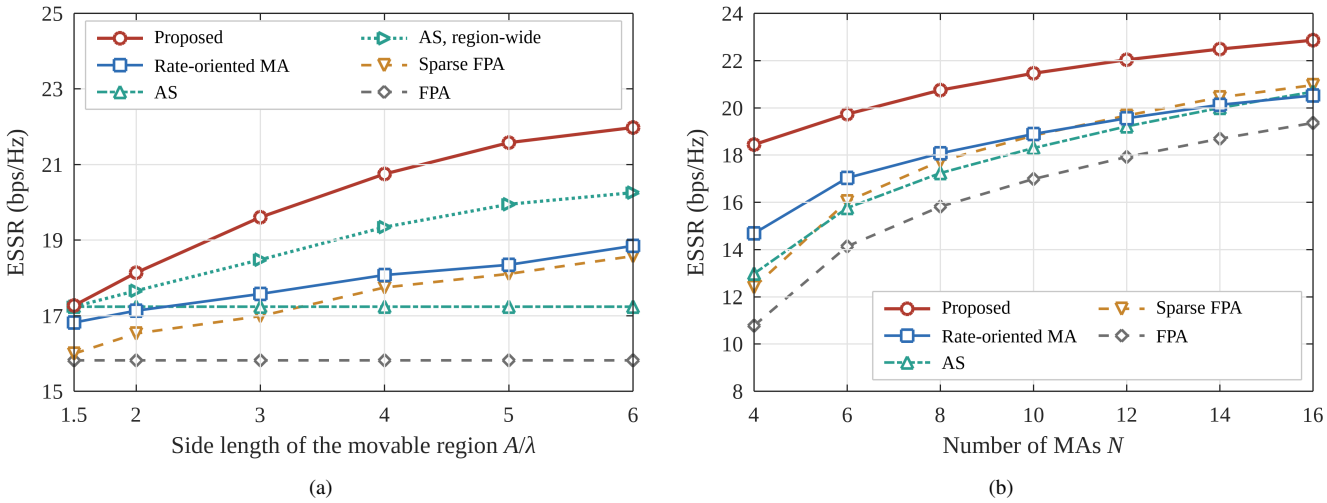


Fig. 6. ESSR versus (a) the side length A of the movable region and (b) the number of MAs N.

the large aperture. At high SNR, the absolute ESSR gains of the proposed design approach constants, since the antenna positions improve the constant term rather than the slope of the ESSR (see Section III-D). Without AN, i.e., with $q = 0$ in Algorithm 2, the ESSR of the proposed design saturates at high SNR, because the SINRs of the LUs and Eve then grow at the same rate with the transmit power [4].

Fig. 5(b) shows that the optimized AN power fraction increases with the SNR and saturates, as predicted by (33). The proposed design allocates much less power to the AN than all the benchmarks, e.g., an AN power fraction of 0.27 versus 0.47 for the FPA at 20 dB. The AN power fractions of the other benchmarks are comparable to that of the FPA. This difference is consistent with the analysis in Section III-D: because the proposed positions reduce the information leakage and increase the AN received by Eve, less AN suffices, and more power remains for the confidential signals.

E. Impact of the Region Size and the Number of MAs

Fig. 6(a) shows the ESSR versus the side length A of the movable region. Here, we also consider AS with region-wide candidates, which form a 4×4 UPA spanning $\mathcal{C}$, whereas the FPA and AS with the compact candidate array do not depend on A. The gain of the proposed design grows with A, because a larger region provides more spatial degrees of freedom for decorrelating the LoS components of Eve and the LUs. The ESSRs of the rate-oriented MA and the sparse FPA increase much more slowly: the former exploits the enlarged region only to decorrelate the LUs, and the latter cannot adapt its positions to the AoDs. AS with region-wide candidates also benefits from the enlarged aperture and its Eve-aware selection metric, but the proposed design outperforms it with half the number of antennas, because the MAs can be placed anywhere in $\mathcal{C}$ rather than at fixed candidate positions. AS with the compact candidate array performs comparably to the proposed design only for the smallest region, which this array covers entirely.

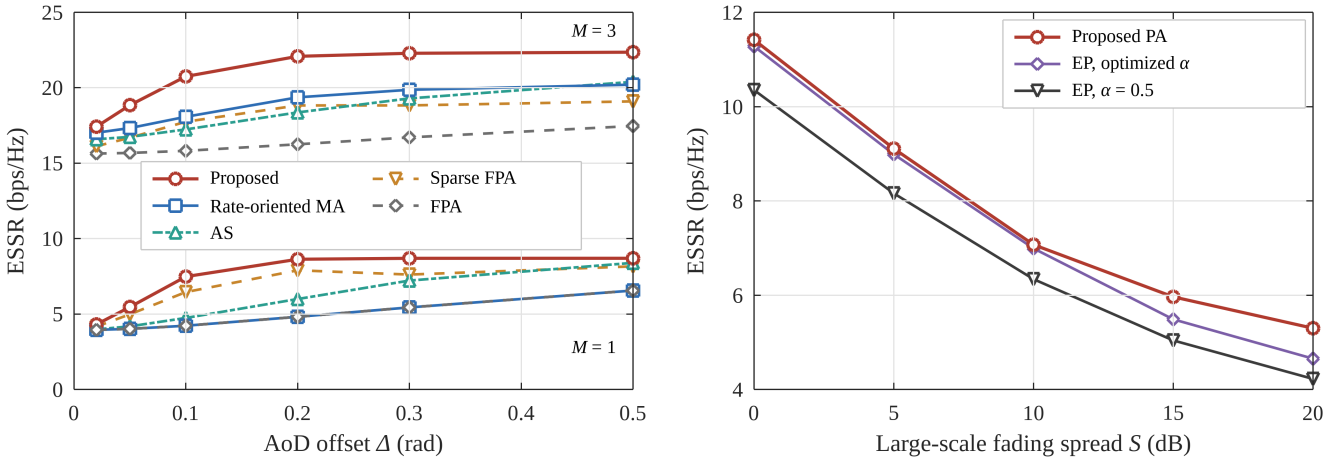


Fig. 7. (a) ESSR versus the AoD offset Δ between Eve and LU 1 for $M = 3$ and $M = 1$, where the rate-oriented MA coincides with the FPA for $M = 1$. (b) ESSR versus the large-scale fading spread S of the LUs for different power allocation (PA) schemes at $P_{\mathrm{tot}}/\sigma^2 = 10$ dB.

Fig. 6(b) shows the ESSR versus the number of MAs N. For $N \geq 12$, the exhaustive search of AS becomes prohibitive, e.g., $\binom{32}{16} \approx 6.0 \times 10^8$ subsets for $N = 16$. In this case, AS adopts a swap search with the same selection metric, which starts from the subset obtained by greedily removing candidate antennas and from four random subsets, and exchanges one selected and one unselected antenna as long as the metric increases. For $N \leq 10$, this search attains the maximum metric of the exhaustive search in at least 92% of the AoD realizations, and the resulting average ESSR differs from that of the exhaustive search by less than 0.03%. All the schemes benefit from more antennas, which provide a larger array gain and more dimensions for the AN. The gains of the proposed design over the FPA and AS decrease with N, because the apertures of their arrays grow with N and allow them to separate Eve from LU 1 better even without position optimization. Nevertheless, the proposed design with $N = 4$ MAs outperforms the FPA with $N = 12$ antennas, and that with $N = 8$ MAs outperforms the rate-oriented MA with $N = 16$ MAs. The secrecy-aware positioning can thus substantially reduce the number of antennas and RF chains required for a target ESSR. The sparse FPA, whose grating lobes are strong for small N, outperforms the rate-oriented MA for $N \geq 14$, because more antennas in $\mathcal{C}$ suppress these lobes.

F. Impact of the AoD Offset

Fig. 7(a) shows the ESSR versus the AoD offset Δ for $M = 3$ and $M = 1$, which determines how close Eve is to LU 1 in direction. For $M = 3$, all the schemes benefit from a larger Δ, because Eve becomes more separable from LU 1. When Eve is almost aligned with LU 1, the proposed design outperforms the rate-oriented MA only slightly, since even an aperture of 4λ can hardly decorrelate their LoS components. As Δ increases, the gain of the proposed design first increases and then decreases, because Eve eventually becomes separable even with Eve-unaware positions. For $M = 1$, the objective of the rate-oriented MA does not depend on the antenna positions (see Section III-D), so that it keeps its UPA initialization and coincides with the FPA. The proposed design, by contrast, substantially improves the ergodic secrecy rate by reducing the LoS correlation $|\bar{\mathbf{h}}_1^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ in (30) and (31), which confirms that the MAs enhance secrecy even for a single LU. The sparse FPA also benefits from its large aperture for $M = 1$, but the proposed design achieves a further gain by adapting the positions to the AoDs.

G. Benefit of the Long-Term Power Allocation

In the default setup, the LUs have identical large-scale fading coefficients and thus similar effective channel gains. To evaluate the long-term power allocation for heterogeneous LUs, we set $[\beta_1, \beta_2, \beta_3] = [0, -S/2, -S]$ dB, where S is the large-scale fading spread, i.e., the ratio of the largest to the smallest large-scale fading coefficient of the LUs in decibels, whereas $\beta_{\mathrm{e}} = 0$ dB. Hence, LU 1, whose direction is close to that of Eve, is the strongest LU, and the other LUs become weaker than Eve as S increases. The proposed power allocation, which combines the secrecy water-filling in Proposition 2 with the optimized AN power, is compared with two equal-power (EP) benchmarks. In these benchmarks, $p_m = \alpha P_{\mathrm{tot}}/M$ and $q = (1-\alpha)P_{\mathrm{tot}}/(N-M)$, where α is either optimized by a one-dimensional search [5] or fixed to 0.5 [14]. For all the schemes, the antenna positions are optimized by Algorithm 2. Since the power of an individual LU cannot be set to zero under EP, the operator $[\cdot]^+$ is retained in the objective of the EP benchmarks, and the gradient in Stage 2 is computed over the LUs with positive secrecy rates.

Fig. 7(b) shows the ESSR versus S at $P_{\mathrm{tot}}/\sigma^2 = 10$ dB. The ESSRs of all the schemes decrease with S, because LUs 2 and 3 become weaker. EP with optimized α performs close to the proposed power allocation for $S \leq 10$ dB, since the LUs then have similar effective channel gains, which the optimized positions balance further. As S increases, the gain of the proposed power allocation grows, e.g., to 14%

at $S = 20$ dB, because it allocates less or zero power to the LUs whose effective channel gains are comparable to or lower than those of Eve (see Proposition 2). For $S = 20$ dB, this gain increases to 40% at $P_{\mathrm{tot}}/\sigma^2 = 0$ dB and decreases to below 1% at 20 dB, because all the LUs achieve positive secrecy rates at high SNR, where nearly equal powers become optimal, as in classic water-filling. EP with $\alpha = 0.5$ performs worst for all the values of S, which shows the importance of optimizing the power splitting between the confidential signals and the AN. Therefore, both components of the long-term power allocation matter: the information–AN power splitting improves the ESSR for all the considered values of S, and the per-user power allocation further benefits heterogeneous LUs at low-to-moderate SNR.

VI. Conclusion

This paper has investigated two-timescale secure transmission for an MA-enabled multiuser downlink with a passive Eve. The derived closed-form approximation of the ergodic wiretap rate has revealed that, besides decorrelating the LUs, the MAs enhance secrecy by reducing the information leakage and increasing the AN received by Eve, even for a single LU. Based on this approximation, we have developed an AO algorithm, which combines secrecy water-filling with a two-stage position update and whose objective value is guaranteed to converge. Simulation results have shown that the proposed design outperforms the rate-oriented MA design, AS, and FPA arrays, while reducing the AN power and the number of antennas required for secure transmission. Extending this design to multiple multi-antenna eavesdroppers and imperfect CSI of the LUs is left for future work.

Appendix A
Proof of Lemma 1

Let $\mathbf{U} \in \mathbb{C}^{N\times(N-M)}$ have orthonormal columns that span the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$. Then, $\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{U}\mathbf{U}^{\mathrm{H}}$ and $Y = \|\mathbf{U}^{\mathrm{H}}\mathbf{g}(\mathbf{t})\|^2$. Since $\bar{\mathbf{w}}_m$ is a unit-norm vector in the column space of $\mathbf{H}(\mathbf{t})$, $\tilde{\mathbf{U}} \triangleq [\bar{\mathbf{w}}_m, \mathbf{U}]$ has orthonormal columns. Because $\tilde{\mathbf{g}}$ is independent of $\mathbf{H}(\mathbf{t})$, (3) yields $\mathbf{g}(\mathbf{t})|\mathbf{H}(\mathbf{t}) \sim \mathcal{CN}(\sqrt{\kappa_{\mathrm{e}}\tilde{\beta}_{\mathrm{e}}}\,\bar{\mathbf{g}}(\mathbf{t}), \tilde{\beta}_{\mathrm{e}}\mathbf{I}_N)$. Consequently, $\tilde{\mathbf{U}}^{\mathrm{H}}\mathbf{g}(\mathbf{t})$ is conditionally Gaussian with covariance matrix $\tilde{\beta}_{\mathrm{e}}\tilde{\mathbf{U}}^{\mathrm{H}}\tilde{\mathbf{U}} = \tilde{\beta}_{\mathrm{e}}\mathbf{I}_{N-M+1}$, so that its entries are conditionally independent. As a result, $X_m = |\bar{\mathbf{w}}_m^{\mathrm{H}}\mathbf{g}(\mathbf{t})|^2$ and $Y = \|\mathbf{U}^{\mathrm{H}}\mathbf{g}(\mathbf{t})\|^2$ are conditionally independent. Finally, (22) and (23) follow from applying $\mathbb{E}\{|z|^2\} = |\mu|^2 + \tilde{\beta}_{\mathrm{e}}$ and $\mathrm{Var}\{|z|^2\} = \tilde{\beta}_{\mathrm{e}}^2 + 2|\mu|^2\tilde{\beta}_{\mathrm{e}}$ for $z \sim \mathcal{CN}(\mu, \tilde{\beta}_{\mathrm{e}})$ to the entries of $\tilde{\mathbf{U}}^{\mathrm{H}}\mathbf{g}(\mathbf{t})$, together with $|\bar{\mathbf{w}}_m^{\mathrm{H}}\bar{\mathbf{g}}(\mathbf{t})|^2 = \bar{X}_m$ and $\|\mathbf{U}^{\mathrm{H}}\bar{\mathbf{g}}(\mathbf{t})\|^2 = \bar{Y}$.

Appendix B
Derivation of (25) and (26) and Proof of Proposition 1

For brevity, the argument $\mathbf{t}$ is omitted. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are invariant to the scaling of the columns of $\mathbf{H}$, we assume $\mathbf{B} = \mathbf{I}_M$ without loss of generality, so that $\mathbf{H} = \mathbf{L} + \tilde{\mathbf{H}}$ by (17), and $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are given by (12) and (13), respectively. We also use $\mathbb{E}\{\mathbf{H}^{\mathrm{H}}\mathbf{H}\} = \mathbf{\Phi}$, $\mathbb{E}\{\tilde{\mathbf{H}}\} = \mathbf{0}$, and $\mathbb{E}\{\tilde{\mathbf{H}}\mathbf{x}\mathbf{x}^{\mathrm{H}}\tilde{\mathbf{H}}^{\mathrm{H}}\} = \|\mathbf{x}\|^2\mathbf{I}_N$ for any deterministic $\mathbf{x}$.

Derivation of (25)*:* Replacing $(\mathbf{H}^{\mathrm{H}}\mathbf{H})^{-1}$ in the unnormalized ZF direction by $\mathbf{\Omega} = \mathbf{\Phi}^{-1}$ yields $\bar{\mathbf{w}}_m \approx \mathbf{H}\mathbf{\Omega}\mathbf{e}_m/\|\mathbf{H}\mathbf{\Omega}\mathbf{e}_m\|$. Then, we approximate the mean of $|\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{H}\mathbf{\Omega}\mathbf{e}_m|^2/\|\mathbf{H}\mathbf{\Omega}\mathbf{e}_m\|^2$ by the ratio of the means of its numerator and denominator, as in [18, Lemma 1]. These are two separate approximations, since $\mathbb{E}\{(\mathbf{H}^{\mathrm{H}}\mathbf{H})^{-1}\} \neq \mathbf{\Omega}$ in general. Since $\mathbf{H} = \mathbf{L} + \tilde{\mathbf{H}}$ and $\mathbf{L}^{\mathrm{H}}\bar{\mathbf{g}} = \mathbf{v}$, we have

$$\begin{aligned}\mathbb{E}\{|\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{H}\mathbf{\Omega}\mathbf{e}_m|^2\} &= |\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{L}\mathbf{\Omega}\mathbf{e}_m|^2 + \|\bar{\mathbf{g}}\|^2\|\mathbf{\Omega}\mathbf{e}_m\|^2 \\ &= \left|[\mathbf{\Omega}\mathbf{v}]_m\right|^2 + N[\mathbf{\Omega}^2]_{m,m}, \qquad (59)\\ \mathbb{E}\{\|\mathbf{H}\mathbf{\Omega}\mathbf{e}_m\|^2\} &= \mathbf{e}_m^{\mathrm{T}}\mathbf{\Omega}\mathbf{\Phi}\mathbf{\Omega}\mathbf{e}_m = \omega_m, \qquad (60)\end{aligned}$$

whose ratio is given by (25).

Derivation of (26)*:* Replacing $(\mathbf{H}^{\mathrm{H}}\mathbf{H})^{-1}$ in $\mathbf{I}_N - \mathbf{P}^{\perp}(\mathbf{t})$ by $\mathbf{\Omega}$ and using $\mathbb{E}\{\tilde{\mathbf{H}}\mathbf{\Omega}\tilde{\mathbf{H}}^{\mathrm{H}}\} = \mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N$ yield

$$\mathbb{E}\{\mathbf{I}_N - \mathbf{P}^{\perp}(\mathbf{t})\} \approx \mathbf{\Pi} + \mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N, \qquad \mathbf{\Pi} \triangleq \mathbf{L}\mathbf{\Omega}\mathbf{L}^{\mathrm{H}}, \quad (61)$$

where $\mathbf{\Pi}$ and $\mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N$ represent the parts of the signal subspace induced by the LoS and NLoS components, respectively. However, (61) lets the NLoS component capture the fraction $\mathrm{tr}(\mathbf{\Omega})$ of every direction, including those captured by the LoS component, so that $\bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{I}_N - \mathbf{\Pi} - \mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N)\bar{\mathbf{g}}$ may become negative when $\bar{\mathbf{g}}$ is closely aligned with the column space of $\mathbf{L}$. To avoid this double counting, we let the NLoS component capture a fraction $\chi \in [0,1]$ of the complement of $\mathbf{\Pi}$ only, i.e.,

$$\mathbb{E}\{\mathbf{I}_N - \mathbf{P}^{\perp}(\mathbf{t})\} \approx \mathbf{\Pi} + \chi(\mathbf{I}_N - \mathbf{\Pi}). \quad (62)$$

The fraction χ is determined by the identity $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$, which holds for every realization of $\tilde{\mathbf{H}}$. Since $\mathrm{tr}(\mathbf{\Pi}) = \mathrm{tr}\big(\mathbf{\Omega}(\mathbf{\Phi} - N\mathbf{I}_M)\big) = M - N\mathrm{tr}(\mathbf{\Omega})$, taking the trace of (62) yields $M - N\mathrm{tr}(\mathbf{\Omega}) + \chi\big(N - M + N\mathrm{tr}(\mathbf{\Omega})\big) = M$, i.e., $\chi = N\mathrm{tr}(\mathbf{\Omega})/(N - M + N\mathrm{tr}(\mathbf{\Omega}))$. Therefore, $\mathbb{E}\{\mathbf{P}^{\perp}(\mathbf{t})\} \approx (1-\chi)(\mathbf{I}_N - \mathbf{\Pi})$, which yields (26) due to $\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{\Pi}\bar{\mathbf{g}} = \mathbf{v}^{\mathrm{H}}\mathbf{\Omega}\mathbf{v}$.

Proof of Proposition 1: For the range, the matrix inversion lemma yields $\mathbf{I}_N - \mathbf{\Pi} = \mathbf{I}_N - \mathbf{L}(\mathbf{L}^{\mathrm{H}}\mathbf{L} + N\mathbf{I}_M)^{-1}\mathbf{L}^{\mathrm{H}} = (\mathbf{I}_N + \mathbf{L}\mathbf{L}^{\mathrm{H}}/N)^{-1} \succ \mathbf{0}$. Together with $\mathbf{\Pi} \succeq \mathbf{0}$, this implies $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$. Moreover, $\mathbf{\Phi} \succeq N\mathbf{I}_M$ implies $0 < \mathrm{tr}(\mathbf{\Omega}) \leq M/N$, i.e., $1 - \chi \in [1 - M/N, 1)$. Hence, $d = (1-\chi)\bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{I}_N - \mathbf{\Pi})\bar{\mathbf{g}} \in [0, N]$ due to $\|\bar{\mathbf{g}}\|^2 = N$. For c_m, the Cauchy–Schwarz inequality yields $|[\mathbf{\Omega}\mathbf{v}]_m|^2 = |\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{L}\mathbf{\Omega}\mathbf{e}_m|^2 \leq N\mathbf{e}_m^{\mathrm{T}}\mathbf{\Omega}\mathbf{L}^{\mathrm{H}}\mathbf{L}\mathbf{\Omega}\mathbf{e}_m$, so that the numerator of (25) is at most $N\mathbf{e}_m^{\mathrm{T}}\mathbf{\Omega}(\mathbf{L}^{\mathrm{H}}\mathbf{L} + N\mathbf{I}_M)\mathbf{\Omega}\mathbf{e}_m = N\omega_m$ due to $N \geq 1$, i.e., $c_m \in [0, N]$. The exact counterparts also lie in $[0, N]$, since $\bar{X}_m, \bar{Y} \leq \|\bar{\mathbf{g}}\|^2$ for the unit-norm $\bar{\mathbf{w}}_m$ and the projector $\mathbf{P}^{\perp}(\mathbf{t})$.

If $\kappa_m = 0$, $\forall m \in \mathcal{M}$, then $\mathbf{L} = \mathbf{0}$, $\mathbf{H} = \tilde{\mathbf{H}}$, $\mathbf{\Phi} = N\mathbf{I}_M$, and $\mathbf{\Pi} = \mathbf{0}$, so that (25) and (26) reduce to $c_m = 1$ and $d = N - M$. In this case, the distribution of $\tilde{\mathbf{H}}$ is invariant to left multiplication by any unitary matrix, which is inherited by $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$. Hence, $\bar{\mathbf{w}}_m$ is uniformly distributed on the unit sphere, which yields $\mathbb{E}\{\bar{X}_m\} = \|\bar{\mathbf{g}}\|^2/N = 1$, and $\mathbb{E}\{\mathbf{P}^{\perp}(\mathbf{t})\}$ commutes with every unitary matrix and thus equals $(1 - M/N)\mathbf{I}_N$ due to $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$, which yields $\mathbb{E}\{\bar{Y}\} = N - M$.

Finally, let $\mathbf{K} = \kappa\mathbf{K}_0$ for a fixed diagonal matrix $\mathbf{K}_0 \succ \mathbf{0}$, so that $\mathbf{L} = \sqrt{\kappa}\mathbf{L}_0$ with $\mathbf{L}_0 \triangleq \bar{\mathbf{H}}\mathbf{K}_0^{1/2}$, which has full column rank if the LoS components of the LUs are linearly

independent. As $\kappa \to \infty$, $\mathbf{H}/\sqrt{\kappa} \to \mathbf{L}_0$ almost surely. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are invariant to the scaling of $\mathbf{H}$ and continuous at full-column-rank matrices, and $\bar{X}_m$ and $\bar{Y}$ are bounded by N, the dominated convergence theorem yields $\mathbb{E}\{\bar{X}_m\} \to |\bar{\mathbf{g}}^{\mathrm{H}}\bar{\mathbf{w}}_m^{\infty}|^2$ and $\mathbb{E}\{\bar{Y}\} \to \bar{\mathbf{g}}^{\mathrm{H}}\mathbf{P}_{\infty}^{\perp}\bar{\mathbf{g}}$, where $\bar{\mathbf{w}}_m^{\infty} \triangleq \mathbf{L}_0(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{e}_m/\|\mathbf{L}_0(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{e}_m\|$ and $\mathbf{P}_{\infty}^{\perp} \triangleq \mathbf{I}_N - \mathbf{L}_0(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{L}_0^{\mathrm{H}}$. On the other hand, $\kappa\mathbf{\Omega} \to (\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}$, $\mathbf{\Pi} \to \mathbf{I}_N - \mathbf{P}_{\infty}^{\perp}$, and $\mathrm{tr}(\mathbf{\Omega}) \to 0$. Hence, d converges to $\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{P}_{\infty}^{\perp}\bar{\mathbf{g}}$, and c_m converges to $|[(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{L}_0^{\mathrm{H}}\bar{\mathbf{g}}]_m|^2/[(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}]_{m,m} = |\bar{\mathbf{g}}^{\mathrm{H}}\bar{\mathbf{w}}_m^{\infty}|^2$, since $N[\mathbf{\Omega}^2]_{m,m}$ decays as $\mathcal{O}(\kappa^{-2})$ whereas ω_m decays as $\mathcal{O}(\kappa^{-1})$. This completes the proof.

APPENDIX C
DERIVATIVES OF ω_m, c_m, AND d

Let $\tau_m \triangleq [\mathbf{\Omega}^2]_{m,m}$ and $\tilde{\mathbf{v}} \triangleq \mathbf{\Omega}\mathbf{v}$. Since $\mathbf{\Omega}$ is Hermitian, (52) can be rewritten as $\dot{\mathbf{\Omega}} = -(\mathbf{\Omega}\dot{\boldsymbol{\ell}}_n)(\mathbf{\Omega}\boldsymbol{\ell}_n)^{\mathrm{H}} - (\mathbf{\Omega}\boldsymbol{\ell}_n)(\mathbf{\Omega}\dot{\boldsymbol{\ell}}_n)^{\mathrm{H}}$. Hence, the derivatives of ω_m, τ_m, and $\mathrm{tr}(\mathbf{\Omega})$ with respect to u_n are given by

$$\dot{\omega}_m = [\dot{\mathbf{\Omega}}]_{m,m} = -2\mathrm{Re}\big\{[\mathbf{\Omega}\dot{\boldsymbol{\ell}}_n]_m[\mathbf{\Omega}\boldsymbol{\ell}_n]_m^*\big\}, \tag{63}$$

$$\begin{aligned}\dot{\tau}_m = 2\mathrm{Re}\big\{[\mathbf{\Omega}\dot{\mathbf{\Omega}}]_{m,m}\big\} = -2\mathrm{Re}\big\{&[\mathbf{\Omega}^2\dot{\boldsymbol{\ell}}_n]_m[\mathbf{\Omega}\boldsymbol{\ell}_n]_m^* \\ &+ [\mathbf{\Omega}^2\boldsymbol{\ell}_n]_m[\mathbf{\Omega}\dot{\boldsymbol{\ell}}_n]_m^*\big\},\end{aligned} \tag{64}$$

$$\frac{\partial \mathrm{tr}(\mathbf{\Omega})}{\partial u_n} = \mathrm{tr}(\dot{\mathbf{\Omega}}) = -2\mathrm{Re}\big\{\boldsymbol{\ell}_n^{\mathrm{H}}\mathbf{\Omega}^2\dot{\boldsymbol{\ell}}_n\big\}, \tag{65}$$

where (64) follows from $[\dot{\mathbf{\Omega}}\mathbf{\Omega}]_{m,m} = [\mathbf{\Omega}\dot{\mathbf{\Omega}}]_{m,m}^*$. For c_m in (25), the derivative of $\tilde{\mathbf{v}} = \mathbf{\Omega}\mathbf{v}$ is

$$\dot{\tilde{\mathbf{v}}} = \dot{\mathbf{\Omega}}\mathbf{v} + \mathbf{\Omega}\dot{\mathbf{v}} = -\mathbf{\Omega}\dot{\boldsymbol{\ell}}_n\big(\boldsymbol{\ell}_n^{\mathrm{H}}\tilde{\mathbf{v}}\big) - \mathbf{\Omega}\boldsymbol{\ell}_n\big(\dot{\boldsymbol{\ell}}_n^{\mathrm{H}}\tilde{\mathbf{v}}\big) + \mathbf{\Omega}\dot{\mathbf{v}}, \tag{66}$$

and thus

$$\dot{c}_m = \frac{2\mathrm{Re}\big\{[\tilde{\mathbf{v}}]_m^*[\dot{\tilde{\mathbf{v}}}]_m\big\} + N\dot{\tau}_m - c_m\dot{\omega}_m}{\omega_m}. \tag{67}$$

For $d = (1-\chi)(N - \mathbf{v}^{\mathrm{H}}\mathbf{\Omega}\mathbf{v})$ in (26) with $1-\chi = (N-M)/(N-M+N\mathrm{tr}(\mathbf{\Omega}))$ as in Appendix B, we have

$$\frac{\partial(\mathbf{v}^{\mathrm{H}}\mathbf{\Omega}\mathbf{v})}{\partial u_n} = 2\mathrm{Re}\{\tilde{\mathbf{v}}^{\mathrm{H}}\dot{\mathbf{v}}\} - 2\mathrm{Re}\big\{(\tilde{\mathbf{v}}^{\mathrm{H}}\dot{\boldsymbol{\ell}}_n)(\boldsymbol{\ell}_n^{\mathrm{H}}\tilde{\mathbf{v}})\big\}, \tag{68}$$

$$\begin{aligned}\dot{d} = -(1-\chi)\Bigg(&\frac{\partial(\mathbf{v}^{\mathrm{H}}\mathbf{\Omega}\mathbf{v})}{\partial u_n} \\ &+ \frac{N(N - \mathbf{v}^{\mathrm{H}}\mathbf{\Omega}\mathbf{v})}{N-M+N\mathrm{tr}(\mathbf{\Omega})}\,\frac{\partial \mathrm{tr}(\mathbf{\Omega})}{\partial u_n}\Bigg),\end{aligned} \tag{69}$$

where (68) follows from $\mathbf{v}^{\mathrm{H}}\dot{\mathbf{\Omega}}\mathbf{v} = -2\mathrm{Re}\{(\tilde{\mathbf{v}}^{\mathrm{H}}\dot{\boldsymbol{\ell}}_n)(\boldsymbol{\ell}_n^{\mathrm{H}}\tilde{\mathbf{v}})\}$. Substituting (63), (67), and (69) into (53) yields $\nabla_{\mathbf{t}_n}F$. All the quantities in (63)–(69) follow from matrix-vector products with $\mathbf{\Omega}$, requiring $\mathcal{O}(M^2)$ operations.

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